INVESTIGATING THE DYNAMICS AND DECOHERENCE OF A JAYNES-CUMMINGS MODEL (JCM)

DEPARTMENT OF PHYSICS / FACULTY OF SCINCE
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INVESTIGATING THE DYNAMICS AND DECOHERENCE OF A JAYNES-CUMMINGS MODEL (JCM)

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ABSTRACT

Quantum communication is a cutting-edge technology which enables for more secure and efficient communication compared to conventional supercomputer. This gives rise to new technology such as quantum internet, quantum cryptography, quantum memory and quantum sensors. Unfortunately, quantum systems are susceptible to decoherence and noise. Quantum decoherence is defined as dynamical destruction of quantum coherence or loss of quantum information due to interaction of open quantum system with its surroundings e.g. vacuum, thermal bath. In this research report, a theoretical analysis was carried out for the single atom-field system composed of cavity QED of two-level atom using Jaynes-Cummings Model Hamiltonian. The dissipation due to spontaneous emissions and cavity loss were also investigated using the density matrix formalism and Lindblad master equations. These equations were solved numerically and then simulated using programming to analyze the relationship between atomic population and coherence term with respect to decay constants (γ, κ) over time. We compared these results when there was no decoherence i.e. $(\gamma, \kappa) = 0$ which we expected to have an ideal Rabi Oscillation inside the cavity. Moreover, we analyzed the dynamics of the two-level atom including avoided crossing and collapse and revival phenomena with and without dissipation effect then we observed the dependence on ω_L, ω_0 and λ. Finally, the author also discussed several mitigation strategies to minimize decoherence in a two-level system.

Keywords : Decoherence, dissipation, Jaynes-Cummings Model, Density Matrix Formalism, Master Equations, Cavity QED, dynamics.

[MENYELIDIKI DINAMIK DAN NYAH-KOHERENS MODEL JAYNES-CUMMINGS (JCM)]

ABSTRAK

Komunikasi kuantum merupakan teknologi canggih yang membolehkan komunikasi yang lebih selamat dan cekap berbanding adi komputer konvensional. Ini menghasilkan teknologi baharu seperti internet kuantum, kriptografi kuantum, memori kuantum dan sensor kuantum. Malangnya, sistem kuantum sebegini terdedah kepada nyah-koherens dan kehingaran. Nyah-koherens kuantum ditakrifkan sebagai pemusnahan dinamik koherens kuantum atau kehilangan maklumat kuantum akibat interaksi sistem kuantum terbuka dengan persekitarannya yakni vakum, mandian termal. Dalam laporan penyelidikan ini, analisis teori dijalankan untuk sistem atom-medan tunggal yang terdiri daripada rongga QED sistem dua peringkat menggunakan Model Hamiltonian Jaynes-Cummings. Lesapan akibat pelepasan spontan dan lesapan rongga juga disiasat menggunakan formalisme matriks ketumpatan dan persamaan induk Lindblad (Master Equation). Persamaan ini diselesaikan secara berangka dan kemudian disimulasikan menggunakan pengaturcaraan untuk menganalisis hubungan antara populasi atom dan sebutan koherens dengan pemalar pereputan (γ,κ) seiring waktu. Kita telah membandingkan hasil dapatan apabila nyah-koherens tidak hadir yakni $(\gamma, \kappa) = 0$ dimana kita menjangka ayunan Rabi unggul berlaku didalam rongga. Selain itu, penulis juga telah menganalisis dinamik atom dua peringkat termasuklah silangan terhindar serta fenomena kejatuhan dan kebangkitan dengan dan tanpa kesan lesapan, berikutnya memerhatikan kebergantungannya terhadap ω_L, ω_0 dan λ. Akhir sekali, penulis juga membincangkan beberapa pendekatan mitigasi untuk meminimumkan nyah-koherens dalam sistem atom dua peringkat.

Kata kunci : Nyah-koherens, lesapan, Model Jaynes-Cummings, Formalisme Matriks Ketumpatan, Persamaan Induk, QED Rongga, dinamik.

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It would be rationale to begin my acknowledgement part by thanking Lord for sustaining me with a breath of life till now thus be able to accomplish this Final Year Project (FYP) report in excellent. In just a blink of an eye, it feels surreal that my degree journey is nearly at its end and of course this semester was a wonderful opportunity for me to explore the cutting-edge field of quantum technology. Though a bit of challenge and difficulty faced during completion of this report, but I believe with God's blessings, anything is possible!

I owe a debt of gratitude to my supervisor Prof. Dr. Raymond Ooi Chong Heng for his invaluable supervision and for enlightening me in this field of research. Indeed, I've learnt so much from his profound and broad knowledge and experience in Quantum Technology especially when taking his course last semester that is Quantum Optics & Technology. I am sincerely thankful for his insightful feedback on my derivation and works in Quantum Mechanics especially in quantum decoherence matters which tremendously enriched my understanding in this subject.

Not to forget, I would like to express my heartfelt thanks to all my teachers and lecturers from the very first day at school to the present who had played an important role in shaping my academia journey. Last but not least to my parents who I believe their unwavering support and prayers have always been with me. I am deeply thankful to them for their hard work and dedication that have been enabling me to reach up to this point.

Hopefully, this research project could motivate every one of us to keep exploring the mystery of the universe and never stop learning something new. I truly believe that quantum technology will be our near future, time will prove it!

To my teachers, I presented to you a masterpiece of Tao Te Ching from a legendary philosopher, Lao Tzu taken from verse 27 :

"Therefore, the sage takes care of all people,

Forsaking no one.

He takes care of all things,

Forsaking nothing.

This is called following the light"

- Lao Tzu (Tao Te Ching, verse 27) -

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LIST OF SYMBOLS AND ABBREVIATIONS

EM	:	Electromagnetic
JCM	:	Jaynes-Cummings Model
ODEs	:	Ordinary Differential Equations

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CHAPTER 1 : INTRODUCTION

1.1 Introduction

In a layman term, decoherence is simply defined as loss of coherence or quantum information i.e. its states due to interaction of quantum system with environment. Quantum decoherence, sometimes called dynamical quantum decoherence or environmental-induced decoherence emerges from quantum behaviour into classical behaviour (J.K. Gamble & J.F. Lindner, 2008). This means that decoherence is purely quantum mechanical process different from classical dissipation and fluctuation. In reality, we cannot have a quantum system which is completely isolated from the surroundings or environment, such that interaction or entanglement with environment cannot be avoided whereby the interaction strength becomes stronger with a large number of environmental degrees of freedom (M. Schlosshauer, 2019). The environment here can be modeled using quantum harmonic oscillations, thermal bath and vacuum fluctuations.

We always read from quantum mechanics books that, imagine we have a two-level atom in coherent superposition of excited state, |e> and ground state |g>. In quantum optics, this coherent superposition will lead to what so-called Rabi oscillations of the atom due to driving field such as electromagnetic field i.e. light (C. Gerry & P. Knight, 2004). Rabi oscillation is the periodic exchange of energy between these two states |e> and |g> or in short, the electron oscillates back and forth between these energy states. However, this superposition is difficult to prepare and observe due to decoherence caused mechanisms such as spontaneous emission, photon loss and spin-phase flip. These mechanisms are caused by interaction with environment whereby it 'measures' the quantum states hence effectively selecting only one outcome/states out of two which leads to dynamical destruction of superposition – collapse of wavefunction. Theoretically, the off-diagonal terms in density matrix will vanish as we partial trace over the environment i.e. $Tr_{\varepsilon}(\rho_{eg}) = Tr_{\varepsilon}(\rho_{ge}) = 0$.

What causes these decoherence mechanisms? In cavity Quantum Electrodynamics (QED), the main source of the dissipation and decoherence is photon leakage or loss from the cavity due to imperfect reflectivity of the cavity mirror. The photon can then escape the cavity mirror thus coupling the photon (cavity field) to external radiation mode i.e. environment. Spontaneous emission on the other hand occurs when atom in excited state, |e> interacts with the vacuum electromagnetic field. The vacuum has zero-point fluctuations which plays role as perturbation to the atom. This emitted photon will entangle with vacuum field transferring the energy to the field, hence causing spontaneous emission. Besides that, the spin flip decoherence happens when the spin

states of the quantum particle such as electrons of spin half flips due to interaction with environmental magnetic field or spin baths. The environmental spin will somehow exchange the spin with the system spin causing transition from $|\uparrow\rangle$ to $|\downarrow\rangle$ or vice versa. Hence as we trace over the environment of the density matrix results in the phase coherence loss in superposition leading to spin flips decoherence.

Density matrix serves as an important formalism in open quantum system (G. Kaplanek and E. Tjoa, 2022) where interaction with reservoir or environment is unavoidable. Moreover, density matrix formalism is a powerful tool in describing quantum state of any physical system as it's not only allowing pure states picture but also mixed state(C. Kading & M. Pitschmann). However, the latter is not accessible with description using wavefunction (M. Schlosshauer, 2007). The general form of density matrix is shown below.

$$\rho = \sum_i p_i |\Psi_i\rangle\langle\Psi_i| \tag{1}$$

For pure state, i = 1, $p_i = 1$ hence $\rho = |\Psi_1\rangle\langle\Psi_1|$ represented by Ψ_1. As for mixed state say i = 2, $\rho = \frac{1}{2}|\Psi_1\rangle\langle\Psi_1| + \frac{1}{2}|\Psi_2\rangle\langle\Psi_2|$, we cannot have one wavefunction to represent this.

The paper is divided into 6 chapters whereby the first 2 chapters are as introductory to my research project. Chapter 3 will first introduce the notion of decoherence function and factors of qubit which describes how quantum system loses their coherence through interaction with environment. Chapter 4 will focus on single atom and light system decoherence which the decoherence is susceptible due to spontaneous emission, cavity loss and spin flip dephasing. The simple Hamiltonian model i.e. Jaynes-Cummings is used for 2- and 3-level atoms. Understanding the decoherence mechanism in a single atom will be significant to enhance the quantum memory system. Chapter 5 will display the results and discussion on this research including depiction of graphs and simulations regarding decoherence. The last chapter is to conclude everything in this paper. Please note that the investigation is only based on Markovian approximation where memory effect of the system about its interaction with quantum state is omitted i.e. memory-less evolution.

Qubit system such as quantum memory should be engineered in such a way that minimizes decoherence and dissipation. Of course, we cannot make it completely isolated or closed otherwise we lose control of the system (M. Schlosshauer, 2007). Several ways to control decoherence including Quantum Error Correction (QEC), encoding the quantum information in decoherence-free subspace (DFS), Cavity QED Approach i.e. trapping the atom inside cavity for strong coupling as well as dynamical decoupling method. We will discuss this at the end of chapter 4.

1.2 Background of study

Quantum communication often relies on photon as a quantum information carrier over long-distance. In classical transmission of photons, the fiber optics is used together with classical repeater to overcome loss due to fiber attenuation. An optical repeater works by converting the incoming optical signal (light) to electrical signal for processing. The signal is amplified and reshaped then eventually converting the signal back to optical signal with increased intensity. However, for the case of quantum communication, it is not possible to implement this technique i.e. converting the light into electrical signal for processing because the act of processing itself will 'measure' the light state causing collapse of wavefunction or superposition.

For quantum repeaters, a long-haul communication channel is divided into several segments with nodes in between. A quantum repeater instead works by creating entanglement of states between intermediate nodes (quantum memory) in communication channel, storing the states in quantum memory to synchronize the process. It performs the entanglement swapping at the node where Bell state measurement taking place to entangled node A and B which located at distance. Unlike classical repeater where signal is modulated, quantum repeater does not rely on amplifying the signal but using quantum error correction (entanglement purification) to mitigate decoherence and loss. Nevertheless, quantum memory is susceptible to quantum decoherence due to interaction with environment. For instance, the interaction with uncontrolled external system, atomic or photonic loss, spontaneous emission and spin flips.

The simplest quantum memory scheme one could imagine is indeed single atom memory. Consider a two-level atom, the atom is in ground state |g> coupled to single optical mode field(light) via an electric dipole transition. As light field (probe), resonant with transition is applied to the atom, it gets excited to excited level |e>. The photon is now 'trapped' as a stationary excitation of the atom and the quantum storage is complete. However, due to electric dipole transition too, the photon absorbed by atom also causes the re-emission of the photon almost instantly. Please note that this is not spontaneous emission yet, but instead the Rabi oscillation of atom between energy state. When the quantum memory couples with the vacuum field then spontaneous emission appears which is driving the atom to decay from |e> to |g>.

One approach to keep the photon 'freeze' at higher state for some longer time is by introducing third state – dark state or sometimes called spin state state, |s> which is in general do not couple to the photonic mode containing information we want to store. We now apply a control pulse to bring the electron from |e> to |s>. In this way, the photon can remain trapped as a spin excitation i.e. the information we store converted from photonic to spin excitation at longer time span. To retrieve it

fairly simple by applying control field once again to convert the spin excitation state into photonic state by transferring the atomic state back to |e> and then |g> due to Rabi oscillation.

Nevertheless, this research focuses more on theoretical analysis of decoherence mechanisms mentioned within single-atom quantum memory system. To model the single atom-light system, the simplest model available is Jaynes-Cummings model. The study investigates ranging from 2-level, 3-level and multi-level atomic configuration in cavity QED. Indeed, we need to model the atomic-light system to interact with environment ε using the simple quantum harmonic oscillator or thermal bath. Moreover, in order to model the decoherence mechanisms, the density matrix formalism for open quantum system that is, Lindblad Master equation is used. The Lindblad master equations take the general form as below.

$$\dot{\rho}(t) = -\frac{i}{\hbar}[H, \rho] + \mathcal{L}(\rho) \tag{2}$$

where the first term describes the dynamics of the system and the second term is the decoherence operator or widely known as Liouvillian operator. Once we solve for $\rho(t)$ either numerically or analytically, it will be convenient for us to explore more on the full dynamics including the atomic coherent, population and the number operator. The decoherence function Γ_{mn} as discussed in chapter 3 describes the behaviour of off resonant of the reduces density matrix. We will first demonstrate the purest form of decoherence i.e. the destruction of quantum coherence without decay population and energy is conserved. This is only due to system-reservoir interaction, caused by pure dephasing not spontaneous emission or photon loss. At the end, the author committed to discuss the decoherence mitigation approach for atomic ensemble quantum memory.

1.3 Objective of study

The main objective of the study is to analyze and investigate the dynamics of the two-level atom and field interaction using Jaynes-Cumming Model (JCM). The phenomena discussed in the research including the avoided crossing, damped JCM and collapse and revival. The study has considered these phenomena with and without dissipation mechanisms such as spontaneous emission and photon loss (or cavity loss) and observing the effect of dissipation to the population and coherence in the density matrix elements of the atom-field system. For this task, the study has implemented the Lindblad master equation.

Moreover, the study proposed to conduct simulations to validate the theoretical findings of the dissipation factors $i.e.\gamma$(due to spontaneous emission) and κ(due to cavity decay) using programming.

Apart from that, there are several approaches to mitigate decoherence in quantum memory system to achieve high-performance memory capability. This will be discussed in chapter 6.

1.4 Author's motivation

Quantum Optics is the field of study in physics that has been rapidly growing in recent times. It is the heart of the development of new cutting-edge quantum technology which is now at the forefront of scientific research and development (R&D). I personally believe that quantum technology will be our driving future just like boomer's generation perceive the advancement of recent technology. The conventional internet will be replaced by quantum internet, conventional AI into quantum AI, conventional communication into quantum communication and cryptocurrency into quantum cryptocurrency. As quoted from Eleanor Roosevelt, 'The future belongs to those who believe in the beauty of their dreams'. Indeed, my motivation of doing this research is not merely based on future development of quantum technology!

During last semester (semester 6), I decided to take Quantum Optics & Technology (SIF3020) course taught by Prof. Raymond. It was a first starting leap of mine to learn more about Quantum Technology. The professor taught the course quite well as I know this is his field of expertise, so no wonder. With his own style of teaching explaining the theory and formula of quantum optics somehow amazed in some way. If you are taught by him, never miss asking questions to strengthen the knowledge in some topics or subtopic. I admitted that quantum optics was a new course for me, and following his lecture every time didn't guarantee to understand the subjects pretty well. So I asked him, in several occasions! Moreover, through his teaching methods and explanation, I've learnt an important skill to be a good physicist i.e. a good understanding of theory and formula.

Hence, I made up my mind in choosing quantum optics field as my final year project (FYP) and Prof. Raymond as my supervisor. I am eager to dive into the decoherence mechanisms of quantum system such as quantum memory. Through intensive readings, I found out that decoherence is the major issue in developing an efficient quantum system. So why not to give it a try on discovering it, isn't? In conclusion, I sincerely hope that one day I will have the opportunity to contribute on this state-of-art quantum technology.

CHAPTER 2 : LITERATURE REVIEW

2.1 Quantum Optics

Quantum optics is the modern branch of the field in physics. The quantum description of light that produces nonclassical and entangled effects has revolutionized a new perspective in optics. Ironically, the results obtained from quantum theory involving interaction between two particles differs from the results obtained classically. The study of interaction of photon with matters has led to development of quantum optics field (Aini Syahida, 2013). The foundation of quantum optics can be traced back through the work of geniuses such as Planck as a pioneer, Einstein, Paul Dirac, Wigner, Bose, Jaynes, Cummings and many more (G.S Agarwal, 1974). The pioneer, Max Planck worked on resolving the ultraviolet catastrophe (M. Planck, 1902) became the building block of quantum optics and quantum mechanics overall. The solution proposed by him worked out very well by simply quantizing the photon of energy called quanta. Besides, the photoelectric effect as proposed by Albert Einstein has shown the particle nature of light as well as his work in A and B coefficients to describe radiation-matter interaction (A.Einstein, 1905). Not forgetting Dirac's work in quantization of electromagnetic(EM) field (P.A.M Dirac, 1927) as well as Weisskopf's and Wigner's work on decay of the excited state which shown to be an exponential decay (V. Weisskopf & E. Wigner, 1930). The heart of quantum optics is Jaynes-Cummings model which is a theoretical quantum framework describing the interaction between quantized field and atomic system. It is the most simplistic quantum model to understand light-matter interaction (E.Jaynes & F.Cummings, 1963). EM field quantization is when we treat the light source as collection of discrete energy packet. It arises upon applying quantum mechanics to the EM field that is by representing the field as harmonic oscillator at each point in space where the oscillation modes correspond to photon's state. The field quantization employs annihilation (a) and creation (a^+) operators which lowers down or rises up the photon number, allowing for solid description of phenomena such as single-photon states, coherent states and squeezed state (Glauber, 1963).

2.2 Quantum decoherence

The attempt to transmit quantum information in the form of light by direct transmission will reduce the rate of communication exponentially with distance due to coherence and loss (J. B. Brask & A. S. Sørensen, 2008). The use of quantum repeater together with quantum memory is important to reduce the effect of decoherence due to interaction with environment. This instead leads to interaction or entanglement between the quantum system with the environment, effectively causing

the loss of information (O. Oreshkov & Todd A. Brun, 2007). The spontaneous emission, cavity loss and pure dephasing and flip are several decoherence mechanisms. Purcell magnum opus paper in 1946 of his work related to magnetic resonance had shown that spontaneous emission as he thought previously as atomic transition from excited to ground state (E.M Purcell, 1946). It was quantum optician dream to confine an atom inside a cavity or resonator where trapped ions or atoms can be strongly coupled or entangled with electromagnetic field inside cavity which then allows the most powerful application in quantum processing and communication (C.C Garry & P. L Knight, 2005). Decoherence simply means when the open quantum system interacts with its reservoir or environment creating correlations between the states of the system and environment as the environment itself contains information to the open quantum open system to form correlations. As soon as we trace over the environment degree of freedom, certain set of states of open system's Hilbert space exhibits a strong stability properties while the superpositions of these states are destroyed instantaneously (H.P Breuer & F.Petruccione).

We can investigate decoherence by using open quantum formalism, one of it is using density matrix given by :

$$\rho = \begin{pmatrix} \rho_{ee} & \rho_{eg} \\ \rho_{ge} & \rho_{gg} \end{pmatrix} \tag{3}$$

The diagonal elements are called the population of the atom in this case 2-level since we have 2 by 2 matrix, and it must obey Tr (ρ) = $\rho_{ee} + \rho_{gg} = 1$ while the off-diagonal elements are called the coherence of the system, and it must obey $\rho_{eg} = \rho_{ge}^{*}$ since the density matrix is Hermitian. From here we can inspect the off-diagonal elements of the density matrix to extract the information about the coherence or superposition of the any pure quantum state (M. Hazra, D. Goswanmi, 2023). The coherence can also be represented by expectation value of the raising and lowering operators of the atom, $\sigma_{\pm}$ which are given by Block equations. Several ways to mitigate decoherence by confining the atom inside a high-finesse cavity.

Mathematically, when we have the wavefunction such that given by,

$$\Psi(t) = \sum_{n} c_n |n\rangle\langle\phi_n(t)| \tag{4}$$

where $\langle\phi_n(t)|$ is an arbitrary reservoir state.

Then the reduced density matrix of the system is given by,

$$\rho_S(t) = tr_B|\Psi(t)\rangle\langle\Psi(t)| = \sum_{m,n} c_n c_m^{*} |n\rangle\langle m|\langle\phi_m(t)|\phi_n(t)\rangle \tag{5}$$

Because of the measurement-type interaction, the reservoir carries all information on the system

state. Due to the measurement-type interaction the reservoir carries information on the system state. But however, the state $|\Psi(t)\rangle$ still superposition involving all states $|n\rangle$. Let us consider extreme case m≠n where the overlap of the states $\langle\phi_m(t)|$ and $\langle\phi_n(t)|$ decrease to zero and eventually causing decoherence with time τ_D. This then leads to reduced density matrix as,

$$\rho_S(t) = \sum_n |C_n|^2 \, |n\rangle\langle n| \quad (6)$$

The coherence of the density matrix in the n basis have lost as a result of interacting or overlapping with the environment.

2.3 Importance of JCM in Quantum Technologies

The application of Jaynes-Cummings Model in quantum technology is apparent as it describes the interaction between light and matter system. The dynamics of the two-level system such as the avoided crossing, Rabi oscillations with and without effect of dissipation as well as atomic inversion which tells us about collapse and revival phenomena can be explored using JCM (W. P. Schleich, 2001).

Moreover, the application in quantum computing such as in Shor's quantum algorithm for factoring large numbers, we need to prepare qubits and quantum logic gates (H. Azuma, 2011). In addition for method of implementation such as interaction between cold trapped ions with laser beams (J. I. Cirac & P. Zoller, 1995) and polarized photons in optical QED cavity (Q.A. Turchette et.al, 1995) require the holistic understanding of JCM system. According to H. Azuma, the preparation of quantum computation in general, follows several procedure i.e. first the preparation of superposition two-state quantum system (qubits) then to apply the unitary transformation using quantum logic gate. Finally, we can obtain an output by observing the qubits using correct measurement basis.

Though JCM or Rabi Model cannot well describe the solid-state CQED system which is usually in the region of Ultra strong coupling (USC) since there are multiple dipoles take part in light-matter system. Hence an extension to JCM model has to be done that is multiple TLS are coupled to the same photonic mode leading to Dicke model (D.D Bernardis et.al , 2024). Hence, an extension to JCM will lead to an application with a multilevel and multimode quantum systems.

Apart from that, in quantum information processing such as quantum repeater, quantum memory and QKD requires deep understanding of fundamental in JCM. In real world, of course the quantum system is susceptible to the effect of decoherence and dissipation. Luckily, JCM can afford to describe the atom-field interaction under the effect of damping such as spontaneous emission and

lossy cavity (J.H. Eberly et al, 1980). In a research had been done by L. Salatino and friends (2023), they were using the JCM to simulate the signal attenuation in an optical fibre for a BB84 QKD scheme. Hence, quantum processors can be used to describe the dynamics of the information transmission process. The conversion of photonic to spin excitation state can be achieved using the two-level atom inside a cavity – The simplest quantum memory scheme. Usually, for a robust used of quantum memory application, the three-level Λ configuration is used instead of two-level system enabling the realization of quantum logics and interfaces between atoms and photons.(A. Karnieli & S. Fan, 2023). Therefore, in this research we are going to explore holistically the whole picture of dynamics of JCM system with and without dissipation effects.

2.4 Theory of spontaneous emission

Spontaneous emission is one of the simplest quantum mechanical processes where an excited atom interacts with quantized EM field (M.V Gurvey, 2011) then transitions back to ground state. Historically, the spontaneous transition rate for atomic transition was first discovered by Einstein in which independent of any external applied EM field in his inspection of thermodynamics equilibrium. While in Lorentz's theory of light and matter, atomic radiation was attributed to the oscillation of atomic electrons. It was impossible to understand why they would radiate at only certain frequencies classically until Bohr theory of hydrogen atom (quantization of matter) came in 1913 (P. L. Knight, 1996). It was crystal clear that from early stage there was slight relation between Einstein's spontaneous emission and Lorentz idea of radiation.

The modern description of spontaneous emission theory was formulated by Dirac in 1927 and then followed by Victor Weisskopf's and Eugene Wigner's work on spontaneous emission in the framework of QED in 1930 (V.Weisskopf & E. Wigner). This work remains the standard of the theory of spontaneous emission till today. Though it provides a good approximation, yet still not rigorously correct when applied in a consistent fashion that maintains unitary as the negative frequencies be permitted in the emitted spectrum is required (P.R. Berman, & G.W. Ford, 2010). Weisskopf and Wigner assumed two assumptions such that the density of states of the vacuum QED field can be evaluated at atomic transition frequency and the frequency is extended over the vacuum field mode to minus infinity in calculating the excited state transition amplitude. An atom in an excited state decay as a result of its interaction with vacuum field which can be described by the solving the amplitude equation as,

$$\dot{b}_2 = -i\sqrt{\frac{\gamma}{\pi}}\int_{-\infty}^{+\infty} f(\omega_k)\, e^{-i(\omega_k-\omega_0)t} b_{1k}, \dot{b}_{1k} = -i\sqrt{\frac{\gamma}{\pi}} f(\omega_k) e^{-i(\omega_k-\omega_0)t} b_2 \qquad (7)$$

where b_2 is the probability amplitude for the atom be in excited state, |e> with no photon (vacuum state), b_{1k} is the probability amplitude for the atom to be in a ground state |g> with a photon having frequency of ω_k, ω_0 is atomic transition frequency, γ is the excited state decay rate and $f(\omega_k)$ is a dimensionless real function that is frequency dependent.

In quantum electrodynamics (QED), the harmonics oscillator with frequency ω_L has a zero-point energy (ZPE) that is $\frac{1}{2}\hbar\omega_L$. This is similar to EM field (P.W. Milonni) – As we quantized the EM field using harmonics oscillator. Sometimes this can be identified as vacuum QED or ground state of the EM field. As a result of the interaction, the electron transitions from |e> to |g> has mixed with the transition of the EM field from vacuum to a higher state field with 1 photon (H. Yokoyama & Ujihara K). Hence the process of spontaneous emission in free space relies upon vacuum fluctuations(M. O Scully & M. S. Zubairy). Spontaneous emission is a fully quantum mechanical process and has no classical analogue.

The ratio of Einstein spontaneous emission coefficient, A and stimulated emission B is given by :

$$\frac{A_{21}}{B_{21}\rho(\omega_0)} = e^{\frac{\hbar\omega_0}{kT}} - 1 \tag{8}$$

where $\rho(\omega_0)$ is the spectral energy density, k is Boltzmann constant and T is temperature.
The rate of atomic transition from |e> to |g> with a frequency of $\omega_0 = \frac{E_e - E_g}{\hbar}$ is given by,

$$R_{21} = B_{21}\rho(\omega_0) = \frac{4\pi d^2}{3\hbar^2}\rho(\omega_0) \tag{9}$$

where d is electric dipole matrix element.
According to this expression, the vacuum QED should induce the atom in the excited state to ground state at a rate of,

$$R_{vf} = \frac{4\pi d^2}{3\hbar^2}\left(\frac{\hbar\omega_0^3}{2\pi^2 c^3}\right) = \frac{1}{2}A \tag{10}$$

This has effectively proven the phenomena of spontaneous emission was induced by vacuum field at a rate equals to half of spontaneous emission rate (P.W. Milonni). But fluctuating field is not the only contribution to spontaneous emission, another factor is the radiation reaction. The rate of oscillation with amplitude x_0 is,

$$\frac{d\varepsilon}{dt} = \left(\frac{e^2\omega_0^2}{3mc^3}\right)\varepsilon_0 \tag{11}$$

where $\varepsilon_0 = m\omega_0^4 x_0^2$ is the energy of the oscillation. Then by replacing $\frac{e^2}{m}$ by $\frac{e^2 f}{m}$ where $f = \frac{2md^2\omega_0}{e^2\hbar}$ is the transition oscillator strength. Hence we will have the radiation reaction as,

$$R_{rr} = \left(\frac{e^2\omega_0^2}{3mc^3}\right)\left(\frac{2md^2\omega_0}{e^2\hbar}\right) = \frac{2d^2\omega_0^3}{3\hbar c^3} = \frac{1}{2}A \tag{12}$$

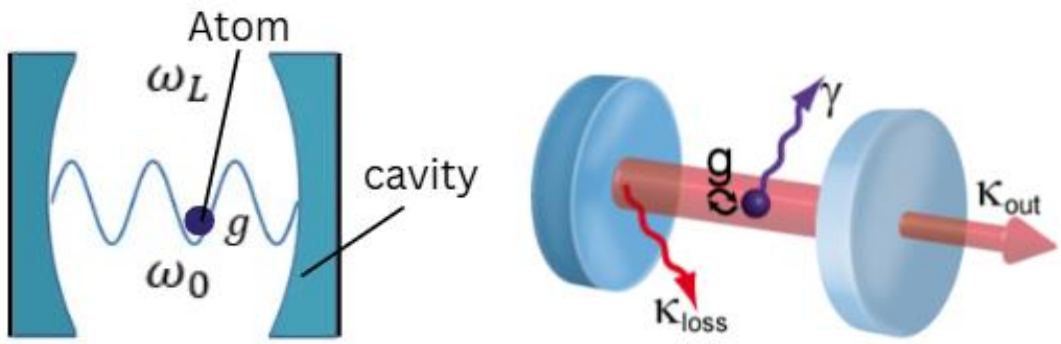
Eventually the total of $R_{vf} + R_{rr}$ will result in the Einstein A coefficient for spontaneous emission, or in other words these contributions cause spontaneous emission.

2.5 Theory of cavity decay

An optical cavity also known as resonator usually consists of a mirror at each end of the elongated part as shown in Figure below. The optical cavity enables light beam to travel back and forth, hence resonating between the mirrors so the amplitude increases very rapidly as it bounces repeatedly (W. T. Silfvas, 2003). The use of slightly curve mirrors can reduce the losses due to refraction and allowing stable beam modes with lower cavity loss. However, in real life, mirrors are not perfectly reflecting the light beam such that some tunneling between cavity mode and external continuum mode still present, resulting in cavity loss or decay (C. Genes, 2019). When we refer to cavity losses actually what it means is the unwanted loss due such as absorption, scattering and interaction of atom with multiple photon modes.

In a research by M. Scala et al, they had explored the microscopic derivation of master equation for JCM with cavity losses where they considered the dissipator terms which vary with frequency of the order of vacuum Rabi frequency i.e. when n = 0. Moreover, C. D Fidio et. al have also done an interesting research on investigating the dynamics of the excited atom in a lossy cavity using quantum trajectory method while the effects on the dynamics of the intracavity modes was treated using quantum Langevin equations or master equations.

However, in our research, we are going to focus more on the dynamical evolution of density matrix of two-level atom using Lindblad master equation for cavity decay. Hence, this will give a better picture of the population as well as coherence matrix element with or without the presence of dissipation and comparing the results.



2.6 The Lindblad Master equation

As mentioned previously, the dynamics of JCM under the effects of dissipation is studied using master equation or specifically Lindblad master equation. The most general form of Markovian master equation is known as Lindblad master equation. By Markovian, we meant the Markovian process, a stochastic process where the future state can only depend on the present state, i.e. memoryless effect. In short, the environment can respond instantaneously to the system changes and does not retain any past interaction with the system (J. Golatkar). In Markovian approximation, we assume that the interaction between the system and environment is very weak. The master equation is just an ordinary differential equation describes the evolution of density matrix for Markovian process.

The closed quantum system is just an ideal version for real system where the occurrence of dissipation such as spontaneous emission and cavity decay can lead to loss of coherence (D. Manzano, 2020). In addition, practically, we cannot avoid the interaction of the system of interest with the environment hence it requires sophisticated method to effectively remove the environment from equation of motion i.e. taking the reduce density matrix. Therefore, the Lindblad master equation is used for this purpose.

The paper presented by by C. A Brasil et.al has derived the Lindblad master equation for zero temperature case. They started from the Born-Markov approximation then to trace out bath degree of freedom (environment) to obtain the Lindblad form. The specific situation is to realize that the Linblads represent the effect on the main system caused by interaction with environment (vacuum or photon modes) and the Markov approximation is the condition to generate the Lindbladian operators (C.A Brasil et al, 2012). In our research, we are only focusing on zero temperature condition for the sake of simplicity. The master equation in Lindblad form for atom-field in k-modes is given by,

$$\dot{\rho}(t) = -\frac{i}{\hbar}[H, \rho(t)] + \sum_k \Gamma_k \left[L_k \rho(t) L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho(t)\} \right] \tag{13}$$

where the first term is the Von Neuman equation describing the dynamics of the unperturbed system and the second term is the dissipator or Liouvillian operator. In contrast, for a single atom interacting with single mode radiation field, the master equation modified as,

$$\dot{\rho}(t) = -\frac{i}{\hbar}[H, \rho(t)] + \Gamma \left[L\rho(t)L^{\dagger} - \frac{1}{2}\{L^{\dagger}L, \rho(t)\}\right] \tag{14}$$

where Γ is decay rate. For spontaneous emission we usually represent it as γ and for cavity decay as κ. Also, the L and $L^{\dagger}$ are the jump operator. For atomic operator we have the σ_{+} and σ_{-} whereas for radiation field, the annihilation (a) and creation ($a^{\dagger}$) operators (H.P. Breuer & F.Petruccione, 2002). As we have discussed about why atom and cavity can decay, i.e. due to interactiong with environment such as vacuum & other photon modes. The complete quantum system for 'atom + vacuum(n=1)' system will be represented by Von Neumann equation and the Hamiltonian used is total 'atom + vacuum state'. As we are interested with the time dependent of the state of the system, the master equation will be (D. Manzano, 2020)

$$\dot{\rho}(t) = -\frac{i}{\hbar}[H, \rho(t)] + \Gamma \left[\sigma_{-}\rho(t)\sigma_{+} - \frac{1}{2}\{\sigma_{+}\sigma_{-}, \rho(t)\}\right] \tag{15}$$

Similarly for the case of cavity decay, the interaction with n-photons mode, the master equation is,

$$\dot{\rho}(t) = -\frac{i}{\hbar}[H, \rho(t)] + \Gamma \left[a\rho(t)a^{\dagger} - \frac{1}{2}\{a^{\dagger}a, \rho(t)\}\right] \tag{16}$$

Take note that the curly bracket refers to anti-commutator i.e. $\{a^{\dagger}a, \rho(t)\} = a^{\dagger}a\rho(t) + \rho(t)a^{\dagger}a$.

CHAPTER 3 : RESEARCH METHODOLOGY

The investigation of the dynamics of 2-level atom using Jaynes-Cummings Model (JCM) will be done fully theoretically and quantitatively. This chapter will describe the theoretical framework, formalism and computational tools used to investigate the dynamics of a two-level atom in a quantized cavity field using JCM.

3.1 The Jaynes-Cummings Model (JCM) Hamiltonian

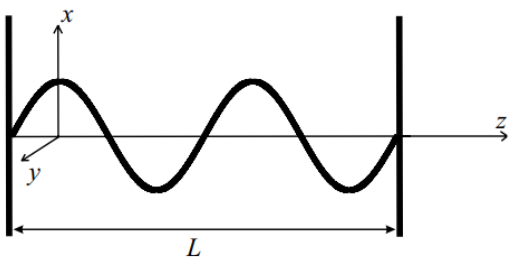
The main ingredient in the theoretical research is the JCM Hamiltonian. JCM is the simplest model in quantum optics to describe the atomic-field interaction. Since the system we are concerned to study is the two-level atom coupled to a single mode of quantized EM field so JCM is the first approach to study the dynamics of the system. Under the RWA, the Jaynes-Cummings Model is usually given by below :

$$H = \frac{1}{2}\hbar\omega_0\sigma_z + \hbar\omega_L a^\dagger a + \hbar\lambda(a\sigma_+ + a^\dagger\sigma_-) \tag{17}$$

where ω_0 and ω_L are the atomic and cavity mode frequency, λ as the coupling strength of dipole transition between excited and ground state, σ_z is Pauli matrix, σ_+, σ_- as atomic raising and lowering operators and $a^\dagger$, a are the creation and annihilation operators.

The Hamiltonian is derived completely in the next chapter and described crystal clear there. One thing to note that the Hamiltonian consists of 3 terms that are atomic term, cavity term and interaction term. The two-level atom is a simple idealized system where we only consider two energy states that are excited (|e>) and ground (|g>) states separated by energy difference of $\hbar\omega_0$.

Next, to describe a fully quantum phenomenon instead of semi-classical (Rabi Model), the Electromagnetic field should also be quantized. Let us begin considering a simple and important case of a radiation field confined to a 1-Dimensional cavity along z-axis such that at z = 0 and z = L it is perfectly conductive walls as shown below :



The crucial tool for the derivation of this term is Maxwell equation,

$$\nabla \times \mathbf{E} = \frac{\delta \mathbf{B}}{\delta t}$$

$$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \frac{\delta \mathbf{E}}{\delta t}$$

$$\nabla \cdot \mathbf{E} = 0$$

$$\nabla \cdot \mathbf{B} = 0 \tag{18}$$

where the quantized fields are given by,

$$E_x(z,t) = \left(\frac{2\omega^2}{V\varepsilon_0}\right)^{\frac{1}{2}} q(t)\sin(kz) \tag{19}$$

$$B_y(z,t) = \frac{\mu_0\varepsilon_0}{k}\left(\frac{2\omega^2}{V\varepsilon_0}\right)^{\frac{1}{2}} \dot{q}(t)\cos(kz) \tag{20}$$

where k is the wave number, V is the confined volume, $q(t)$ is the canonical position and $\dot{q}(t)$ is the canonical momentum of unit mass. These are the basis to form the cavity Hamiltonian term.

After that, we still have the interaction term, which plays a significant role in the investigation. According to F. D. Bonani (2020), the interaction term is the one which unravels the dynamics of the system studied. In JCM, the interaction term is very fundamental to describe the coupling between a two-level atom and a quantized EM field mode. The interaction arises from the dipole interaction which models the interaction between the electric dipole moment of atom and cavity field. A two-level description is valid when we consider the atomic frequency ω_0 are resonant or nearly in resonant with the driving field ω_L i.e. $\omega_0 \approx \omega_L$ (M.O. Scully & M.S.Zubairy, 2012). In short, the interaction of an atom with one of the electromagnetic mode in dipole approximation is given by

$$H_{int} = \frac{e}{mc}\boldsymbol{A} \cdot p_i \approx -\boldsymbol{d} \cdot \boldsymbol{E} \tag{21}$$

where **A** is vector potential, e is electronic charge, m is the mass of electron, c is speed of light and $\boldsymbol{d} = e\boldsymbol{r}$ is the dipole moment of atom with **r** as its position operator.

We can allow certain realistic approximations to reduce the problem to a form which can be solved exactly so that we can extract the essential features of the atom-field interaction. Such approximation such as the Rotating-Wave Approximation (RWA). In RWA, we only keep the slowly varying components in H_{int} beside to keep the conservation of energy such that when the atom absorbs photon of energy, it will transition from |g> to |e> and emitting back the photon of energy when it's transitioning back to lower state. The RWA can be represented as,

$$(\sigma_+ + \sigma_-)(a^\dagger + a) \approx a\sigma_+ + a^\dagger\sigma_- \tag{22}$$

3.2 The use of density matrix formalism

The thorough investigation of the dynamics of JCM deals with the density matrix formalism. Sometimes the density matrix also refers to density operator. The definition of the density matrix is given by below,

$$\rho = \sum_i p_i \, |\psi_i\rangle\langle\psi_i| \tag{23}$$

Of course, in the next chapter, the complete properties and characteristics of ρ is derived. Fortunately, this formalism is capable of describing both pure and mixed states. In describing the non-dissipative system, it is easy to use Schrodinger equation to obtain the population by taking the absolute square of the normalization coefficients since we only have single, coherent quantum state. Nevertheless, for mixed states we cannot do so as we have the representation of many pure states i.e. mixed state. Schrodinger equation assumes the unitary evolution of the pure state while for mixed state, this assumption collapses.

For the case of pure state, we can easily establish the density matrix from the known wavefunction of the system. Besides, for the case of mixed state instead, we can use another approach such as Master equation which accounts for the dissipation and decoherence of the system.

Not only that, what interesting with the density matrix formalism is the measurement of expectation value of any observable, O is simply taking the trace of the observable multiply with the density matrix i.e. $< \hat{O} > = Tr(\hat{O}\rho)$.

3.3 Von Neumann and the master equation

On the other hand, the evolution of density matrix with time can be described using Von Neumann equation. The derivation is carried out in the next chapter using the Schrodinger equation. It is analogous to Schrodinger equation for density matrix as shown below,

$$\dot{\rho}(t) = -\frac{i}{\hbar}[H, \rho(t)] \tag{24}$$

The solution for this is $\rho(t) = e^{-iHt/\hbar}\rho(0)e^{+iHt/\hbar}$ when the Hamiltonian used is independent of time.

Alternatively, we can first take the matrix elements of the density matrix such that we have 4 coupled Ordinary Differential Equations (ODEs), $\dot{\rho_{ee}}(t), \dot{\rho_{eg}}(t), \dot{\rho_{ge}}(t)$ and $\dot{\rho_{gg}}(t)$. While doing so, we need to first calculate the commutator of H and $\rho(t)$.

These ODEs can be solved using python programming using conventional methods such as Laplace transform. After that, we can obtain the atomic population from the elements $\rho_{ee}(t)$ and $\rho_{gg}(t)$ as well as the atomic coherence $\rho_{eg}(t)$ and $\rho_{ge}(t)$. Next the atomic inversion $< \sigma_z >$, can also be calculated which is the difference between $\rho_{ee}(t)$ and $\rho_{gg}(t)$. Apart from obtaining the results, we need to always make sure that our results tally with any relevant literature or sources and verify it according to certain considerations. For instance, we need to make sure that the conditions such as $\rho(t) > 0$ and
$\rho_{gg}(t) + \rho_{ee}(t) = 1$ holds. Also, we need to compare and contrast the results obtained from this research and another research by scholars, indeed with different method and calculation.

To put into consideration the dissipative process such as spontaneous emission and cavity decay/loss, we need to use the master equation as shown below,

$$\frac{d}{dt}\rho(t) = -\frac{i}{\hbar}[H, \rho(t)] + \mathcal{L}_{\text{diss}}[\rho] \tag{24}$$

The first term is the Von Neumann equation while the second is the Liouvillian operator accounts for dissipation. The Liouvillian operator consists of collapse operator, A_- , A_+ which are σ_+, σ_-, $a^\dagger$ and a for this spontaneous and cavity loss case, that is,

$$\mathcal{L}_{\text{diss}}[\rho] = \Gamma\left(A_-\rho(t)A_+ - \frac{1}{2}\{A_+A_-, \rho(t)\}\right) \tag{25}$$

The above expression is general Liouvillian operator where Γ is the rate of dissipation.
Similarly, we need to take the matrix elements of $\dot{\rho}(t)$ since we are interested with the populations and the coherences of atom-field system. Next, since we already have 4 coupled ODEs for the elements, we need to solve it using python programming to obtain the numerical solution for $\rho(t)$. From this we can vary the dissipation rate γ (spontaneous emission) and κ (cavity decay) so that we can keep track the changes in populations plot and how the coherence of the system may be affected. This procedure will be done in the next chapter.

3.4 Numerical and computational method implementation

Due to complex equation obtained from the calculation made, we require an efficient, accurate and effective tool to carry out the calculations. For instance, the solution for coefficient from time-evolution Schrodinger equation, density matrix element from Von Neumann equation and Master equation with dissipative consideration must be solved computationally. The common problem in solving the complex coupled ODEs which has to be solved simultaneously.

We can obtain the solution for coefficients of wavefunction easily using python library called SymPy. The output produced will be in the mathematical form symbolically. The generic procedure is to first define all the parameters symbolically such as the C_e, C_g, C_n, C_{n+1} as the coefficients for the wavefunction, λ, n, ω_0 and ω_L as the coupling strength, photon number, atomic frequency and field frequency respectively. After that, we have to state the initial condition of our coefficients, then to set up the Laplace transform equation then finally to inverse Laplace transform back to obtain the time-domain solutions.

We can do the same procedure for the case of obtaining the symbolical output when solving the coupled ODEs from Von Neumann equation. The solution for density matrix element obtained using Von Neumann equation should be equal to the results calculated using Schrodinger equation. Additionally, we can use the results from Von Neumann equation method to calculate the atomic inversion. After that, we can also plot the results, compare it with the Schrodinger equation method for non-dissipative case as well as to verify if the density matrix element valid using its basic properties studied.

For the dissipative case, it is possible to obtain the full mathematical output from Python programming, unfortunately it takes a long time to run. This is an optimization problem which requires brilliant method to solve but due to insufficient of time, we only managed to plot the results for density matrix elements obtained. Python library called matplotlib.plyplot is a convenient tool to plot any mathematical equation. Firstly, indeed one must solve the four coupled ODEs via Laplace transform then to plot at the end of the code. Differ from solving it symbolically, we have to initialize the value for the parameters $\gamma, \kappa, n, \omega_0$ and ω_L. Interestingly, we can vary the value of the parameters to observe the effects of the dynamics of JCM for non-dissipative and dissipative cases. For instance we can inspect the situation when the values of γ, κ, λ too large or too low, then what will happen to the populations and coherence terms in density operator as well as how it will affect the atomic inversion and photon number variation.

CHAPTER 4 : QUANTUM MECHANICS FORMALISM

4.1 Density Operator

In quantum mechanics, every physical system can be described mathematically using separable complex Hilbert space. That is to say, every quantum state of a system resides in the Hilbert space (A. Galindo P. Pascual, 1989). The state vector or ket contains probabilistic information of the quantum state and is useful to interpret the quantum physical system. Using Dirac notation, quantum state can be represented by ket i.e. $|.\rangle$ and its conjugate form, bra i.e. $\langle.|$. In general, Hilbert space $\mathcal{H}$ is an abstract vector space spans infinite dimension with basis $\{|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, \dots, |\psi_N\rangle\}$ which can we written in wave function form :

$$|\Psi_n(t)\rangle = \sum_{n=1}^{N} \mathrm{C}_i(\mathrm{t})|\psi_N(t)\rangle \tag{26}$$

The wave function is generally obtained by solving the Schrodinger equation as shown below.

$$i\hbar \frac{d}{dt}|\Psi_{\mathrm{n}}(t)\rangle = H|\Psi_{\mathrm{n}}(t)\rangle \tag{27}$$

where H is the time-independent Hamiltonian of the quantum system. Usually, if we are dealing with 2 states (qubit) such as atomic excited $|\mathrm{e}\rangle$ and ground state $|\mathrm{g}\rangle$, we need to solve couple ODEs equation simultaneously. In unitary time-evolution operator, we may write eq. (27) as :

$$|\Psi_n(t)\rangle = U(t)|\Psi_{\mathrm{n}}(t)\rangle = e^{-iHt/\hbar}|\Psi_{\mathrm{n}}(t)\rangle \tag{28}$$

where $U(t) = e^{-iHt/\hbar}$

To describe the ensemble of open quantum system, the density operator or density matrix can be used for this means. It is a language to provide a convenient way for describing quantum system whose its states are completely unknown. In this subsection, we are going to develop an intrinsic characterization of density operators that does not depend on interpretation of ensemble (Michael A. Nielsen & Isaac L. Chuang, 2010). The density operator is then defined by as :

$$\rho(t) = \sum_{i=1}^{N} P_i|\Psi_i(t)\rangle\langle\Psi_i(t)| \tag{29}$$

where $|\Psi_i\rangle$ is the wavefunction of the quantum system and P_i is the probability of the system at the state $|\Psi_i\rangle$. When we have only one value of i or only one term of the density operator, we have the case for pure state such that :

$$\rho = |\Psi_1\rangle\langle\Psi_1| \tag{30}$$

Here we should realize that for pure state, it has single $P_1 = 1$ value.

The density matrix has general properties as depicted below.

1. The trace of ρ equals to 1. (Trace Condition)

$$\begin{aligned} tr(\rho) = \sum_{i=1}^{N} \langle\psi_j|\rho|\psi_j\rangle &= \sum_{i=1}^{N} P_i \langle\psi_j|(|\Psi_i(t)\rangle\langle\Psi_i(t)|)\,|\psi_j\rangle \\ &= \sum_{i=1}^{N} |C_n(t)|^2 \sum_{i=1}^{N} P_i = 1 \end{aligned} \tag{31}$$

where the state vector is assumed to be normalized such that $\langle\Psi_i(t)|\Psi_i(t)\rangle = 1$

2. ρ should always be positive (Positivity Condition)

To show this, suppose we act on the arbitrary vector state $|b\rangle$ and $\langle a|$.

$$\begin{aligned} \langle a|\rho|b\rangle &= \sum_{i=1}^{N} P_i \langle a|\Psi_i(t)\rangle\,\langle\Psi_i(t)|b\rangle \\ &= \sum_{i=1}^{N}\sum_{i=1}^{N} P_i |\mathrm{C}_i(\mathrm{t})|^2 a|\psi_i(t)\rangle\,\langle\psi_i(t)|b\rangle \\ &= \sum_{i=1}^{N} P_i |\mathrm{C}_i(\mathrm{t})|^2 \delta_{a,\psi_i}\delta_{b,\psi_i} > 0 \end{aligned} \tag{32}$$

3. The trace of density matrix square, $Tr(\rho^2)$

$$\begin{aligned} tr(\rho^2) = \sum_{i=1}^{N} \langle\psi_j|\rho^2|\psi_j\rangle &= \sum_{i=1}^{N}\sum_{i=1}^{N} P_i P_j \langle\psi_j|(|\Psi_i\rangle\langle\Psi_i|)^2|\psi_j\rangle \\ &= \sum_{i=1}^{N}\sum_{i=1}^{N}\sum_{j=1}^{N} P_i P_j \langle\psi_j|(|\Psi_i\rangle\langle\Psi_i|)(|\Psi_j\rangle\langle\Psi_j|)|\psi_j\rangle \end{aligned}$$

$$\sum_{i=1}^{N}\sum_{j=1}^{N} P_i P_j \left|\langle\Psi_i(t)|\Psi_j(t)\rangle\right|^2 \leq \left[\sum_{i}^{N} P_i\right] = 1 \tag{33}$$

$$tr(\rho^2) = \begin{cases} 1, & pure\ state \\ < 1, & mixed\ state \end{cases}$$

4. The observable :

$$\langle \hat{O} \rangle = \sum_{i=1}^{N} P_i \langle \Psi_i | \hat{O} | \Psi_i \rangle |$$

$$\langle \hat{O} \rangle = \sum_{i=1}^{N}\sum_{j=1}^{N} P_i \langle \Psi_i | \Psi_j \rangle \langle \Psi_j | \hat{O} | \Psi_i \rangle | = \sum_{i=1}^{N}\sum_{j=1}^{N} P_j \langle \Psi_i | \Psi_j \rangle \langle \Psi_j | \hat{O} | \Psi_i \rangle | = \sum_{i=1}^{N}\sum_{j=1}^{N} P_j \langle \Psi_i | \Psi_j \rangle \langle \Psi_j | \hat{O} | \Psi_i \rangle |$$

$$\langle \hat{O} \rangle = \sum_{i=1}^{N} \langle \Psi_i | \left[\sum_{j=1}^{N} P_j \, | \Psi_i \rangle \langle \Psi_i | \hat{O} \right] | \Psi_i \rangle | = \sum_{i=1}^{N} \langle \Psi_i | \rho O\hat{} | \Psi_i \rangle = Tr(\rho \hat{O}) \tag{34}$$

where the closure relation is $\sum_i |\Psi_i\rangle\langle\Psi_i| = \boldsymbol{I}$. Inserting this relation on the other side of $\hat{O}$, we would obtain :

$$\langle \hat{O} \rangle = Tr(\rho \hat{O}) = Tr(\hat{O} \rho) \tag{35}$$

The matrix representation of the density matrix $\rho(t)$ is given as below :

$$\rho(t) = \begin{pmatrix} \rho_{00}(t) & \rho_{01}(t) & \dots & \rho_{0N}(t) \\ \rho_{10}(t) & \rho_{11}(t) & \dots & \rho_{1N}(t) \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{N0}(t) & \rho_{N1}(t) & \dots & \rho_{NN}(t) \end{pmatrix} \tag{36}$$

The diagonal terms i.e. $\rho_{ii}(t) \in (\mathbb{R} > 0)$ are called populations where as mentioned in (31), the trace must be 1 while the off-diagonal terms i.e. $\rho_{ij}(t) \in \mathbb{C}$ are the coherence of the system which obeys hermiticity.

$$\rho_{ij}(t) = \rho_{jk}^{*}(t) \tag{37}$$

Now imagine a quantum mechanical system consists of two-coupled sub-system as such X and Y. The density matrix of the coupled system is given by ρ_{XY} in the Hilbert space $\mathcal{H}_{XY}$ which is the tensor product $\mathcal{H}_X \otimes \mathcal{H}_Y$. To be clear, X and Y are sub-system respectively while when they coupled/entangled we called it a system. The partial trace of combine density matrix with respect sub-system X will give us the reduced density matrix of Y such that :

$$Tr_X(\rho_{XY}) = \rho_Y \tag{38}$$

The equation above is useful when we have no information of system X but we are intending to study the system Y.

Earlier we discussed the evolution of the wave function from the Schrodinger Equation. Let's find the first derivative of the density matrix from eq. (29).

$$\frac{d}{dt}\rho(t) = \frac{d}{dt}\left[\sum_{i=1}^{N} P_i|\Psi_i(t)\rangle\langle\Psi_i(t)|\right] = \sum_{i=1}^{N} P_i\left[\left(\frac{d}{dt}|\Psi_\mathrm{i}\rangle\right)\langle\Psi_i| + |\Psi_\mathrm{i}\rangle\left(\frac{d}{dt}\langle\Psi_i|\right)\right]$$

$$= \sum_{i=1}^{N} P_i\left[\frac{H}{i\hbar}|\Psi_\mathrm{i}\rangle\langle\Psi_i| + |\Psi_\mathrm{i}\rangle\langle\Psi_i|\frac{H}{-i\hbar}\right] = \frac{-i}{\hbar}(H\rho - \rho H)$$

$$\dot{\rho}(t) = \frac{-i}{\hbar}[H,\rho] \tag{39}$$

We call the equation above Liouville-Von Neumann equation of motion which describes how the density operator evolves with respect to time. It is useful later when we are about to fully describe the dynamics of the quantum system with or without dissipation or decoherence. Further, when the Hamiltonian is time-independent, the solution is given as :

$$\rho(t) = \sum_{i=1}^{N} P_i\,|\Psi_i(t)\rangle\langle\Psi_i(t)| = \sum_{i=1}^{N} P_i\,e^{-iHt/\hbar}|\Psi_i(0)\rangle\langle\Psi_i(0)|e^{+iHt/\hbar} = e^{-iHt/\hbar}\rho(0)e^{+iHt/\hbar} \tag{40}$$

4.2 The Jaynes-Cummings Model (JCM)

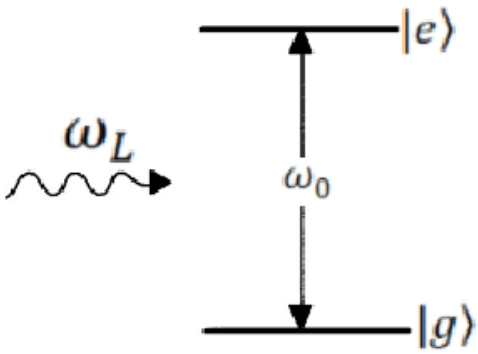
We move our attention to Jaynes-Cumming Model. There is a so called semi-classical, a model of two-level atom interacting with a strong near-resonant classical field which implies the driving field frequency, ω_L matches the atomic frequency, ω_0. But the fully quantum mechanical model to describe this interaction widely known as Jaynes-Cummings model. Specifically, the latter model has no classical analogue which depends wholly on the discreteness of photon and the atomic energy level in this case excited (|e>) and ground (|g>) states as shown below (C.Garry & , 20xx). This model is originally proposed by Edwin Jaynes and Fred Cummings in 1963 which allows us to describe spontaneous emission & cavity loss. (E. Jaynes & F. Cummings, 1963). This model is also capable to find how quantization of field (modelled as harmonic oscillator) differs from the semi-classical model in terms of describing evolution of the two-level atom (F.D Bonani, 2020).

The Hamiltonian of two-level JCM takes the form as follows.

$$H = H_A + H_F + H_I \tag{41}$$

where H_A, H_F, H_I represent Hamiltonian of the atomic energy level, the radiation field and atom-field interaction term.

The full derivation of the JC model is carried out in subsection 4.2.1.



4.2.1 Two-level atom

We begin by exploring the fundamental atom-field interaction of a two-level system. The most basic quantum memory scheme is the two-level atom interacting with photons. Later in this report, we are going to unravel the dissipative and decoherence mechanisms of the two-level atom to understand how quantum memory really works in first principle. We consider the quantum physical system consists of an atom coupled to a model of EM field inside isolated cavity. Say the frequency ω_L of the interested mode of the field then $\hbar\omega_L$ will be approximately equal to energy difference between the |g> and |e>, where only these 2 states have significant in our dynamics for two-level atom. The derivation of JC model begins with,

$$H = \frac{1}{2}\int\left(\varepsilon_0\left|\vec{E}\right|^2 + \frac{1}{\mu_0}\left|\vec{B}\right|^2\right)d^3r + \sum_i\left[\frac{p_i^2}{2m_e} + V(\vec{r_i})\right] - \sum_i\left[\frac{e}{mc}\vec{A}\cdot\vec{p_i}\right] \tag{42}$$

where $\left|\vec{E}\right|, \left|\vec{B}\right|, |\vec{A}|$ are electric field, magnetic field and Coulomb gauge, p_i is electron momentum, m_e is electron mass, e is electron charge and c is speed of light. Realize that this Hamiltonian contains 3 terms which corresponds to $H_F = \frac{1}{2}\int\left(\left|\vec{E}\right|^2 + \left|\vec{B}\right|^2\right)d^3r$, $H_A = \sum_i\left[\frac{p_i^2}{2m_e} + V(\vec{r_i})\right]$ and $H_I - \sum_i\left[\frac{e}{mc}\vec{A}\cdot\vec{p_i}\right]$ where quantization of these terms is required!

Quantization of atom.

The possible atomic states for two-level system is |e> and |g>. It will be convenient to define the state vector (ket) and dual space (bra) of these states in matrix form as such.

$$|e\rangle = \begin{pmatrix}1\\0\end{pmatrix}\ ;\ \langle e| = \begin{pmatrix}1 & 0\end{pmatrix}$$

$$|g\rangle = \begin{pmatrix}0\\1\end{pmatrix}\ ;\ \langle g| = \begin{pmatrix}0 & 1\end{pmatrix} \tag{43}$$

Then, the atomic Hamiltonian can be written as below :

$$H_A = E_e|e\rangle\langle e| + E_g|g\rangle\langle g|$$
$$= \frac{1}{2}(E_e + E_g)(|e\rangle\langle e| + |g\rangle\langle g|) + \frac{1}{2}(E_e - E_g)(|e\rangle\langle e| - |g\rangle\langle g|)$$
$$= \frac{1}{2}(E_e + E_g)I + \frac{1}{2}(E_e - E_g)\sigma_z \tag{44}$$

where I identity matrix defined as $I = |e\rangle\langle e| + |g\rangle\langle g| = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

We suppose that the zero energy of the atom to be halfway between |e> and |g> states. Therefore, we can $(E_e + E_g)$ to be zero energy since we only concern on the energy difference that is $(E_e - E_g) = \hbar\omega_0$. The atomic Hamiltonian then becomes:

$$H_A = \frac{1}{2}\hbar\omega_0\sigma_z \tag{45a}$$

where $\hbar$ is reduced Planck constant and σ_z is Pauli operator given below.

$$\sigma_z = |e\rangle\langle e| - |g\rangle\langle g| = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{45b}$$

$$\hbar = \frac{h}{2\pi}$$

We recall other two Pauli operators which will be useful later,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{46}$$

Quantization of field.

$$H_F = \frac{1}{2}\int \left(\varepsilon_0|\vec{E}|^2 + \frac{1}{\mu_0}|\vec{B}|^2\right) dV \tag{47}$$

The electric and magnetic fields in z-direction are given by,

$$\vec{E}_x(z,t) = \left(\frac{2\omega_L^2}{V\varepsilon_0}\right)^{\frac{1}{2}} q(t)\sin(kz)$$

$$\vec{B}_y(z,t) = \left(\frac{\mu_0\varepsilon_0}{k}\right)\left(\frac{2\omega_L^2}{V\varepsilon_0}\right)^{\frac{1}{2}} \dot{q}(t)\cos(kz) \tag{48}$$

Substituting this into the H_F and integrating it over cavity volume which then simplifies into :

$$H_F = \frac{\varepsilon_0}{2}\int \left(\omega_L^2 q^2(t) + \frac{p^2(t)}{m^2}\right) dV = \frac{1}{2}\left(\omega_L^2 q^2(t) + p^2(t)\right) \tag{49}$$

where $\hat{q}(t) = \left(\frac{\hbar}{2m\omega_L}\right)^{\frac{1}{2}}(\hat{a} + \hat{a}^+)$ is the generalized coordinate and $\hat{p}(t) = -i\left(\frac{\hbar m\omega_L}{2}\right)^{\frac{1}{2}}(\hat{a} - \hat{a}^+)$ is the generalized momentum which satisfies the commutator relation of $[\hat{p}(t), \hat{q}(t)] = i\hbar$ as well as

$$\hat{a} = \sqrt{\frac{m\omega_L}{2\hbar}}(\hat{q} - \frac{i}{m\omega_L}\hat{p})$$

$$\hat{a}^+ = \sqrt{\frac{m\omega_L}{2\hbar}}\left(\hat{q} + \frac{i}{m\omega_L}\hat{p}\right) \tag{50}$$

are annihilation ($\hat{a}$) and creation ($\hat{a}^+$) operators respectively where we let m = 1 for simplicity. Using these variable to substitute into eq.(49) then we get the Hamiltonian of the field as below.

$$H_F = \hbar\omega_L\left(\hat{a}^+\hat{a} + \frac{1}{2}\right) \approx \hbar\omega_L\hat{a}^+\hat{a} \tag{51}$$

The interaction Hamiltonian

We start off by defining the interaction Hamiltonian as,

$$H_I = -\vec{\boldsymbol{d}} \cdot \vec{E} \tag{52}$$

A single atom is coupled to single optical mode via an electric dipole transition or dipole moment operator denoted by $\vec{d}$. The dipole moment which corresponds in atom-field interaction permits the photon to be absorbed by atom as well as re-emit it back almost instantaneously. These two processes occurs back and forth, such that over time the population of excited is oscillating continuously as atoms absorbs and re-emit photon (J.Nunn, PhD thesis 2008). This phenomenon is known as Rabi oscillation, named after physicist Isidor Isaac Rabi who pioneered the theory in the context of nuclear magnetic resonance (I.I. Rabi, 1938). The definition of electric field is,

$$\vec{E} = e\sqrt{\frac{\hbar\omega_L}{V\varepsilon_0}}(\hat{a} + \hat{a}^+)\sin(kz) \tag{53}$$

Eq.(52) then becomes,

$$H_I = \hat{d}g(\hat{a} + \hat{a}^+) \tag{54}$$

where $\hat{d} = \widehat{\boldsymbol{d}} \cdot \boldsymbol{e}$ and $g = -\sqrt{\frac{\hbar\omega_L}{\varepsilon_0 V}}sin(kz)$ is the coupling strength of atom-field interaction.

Let us introduce the atomic transition operator which is the projection of one state to another state and the inversion operator as in eq. (45b).

$$\hat{\sigma}_+ = |e\rangle\langle g|\ ;\ \ \hat{\sigma}_- = |g\rangle\langle e| = \hat{\sigma}_+^+ \tag{55}$$

The operators above obey the commutator relations (Pauli spin algebra) .

$$[\hat{\sigma}_+, \hat{\sigma}_-] = \hat{\sigma}_z\ \ ;\ \ [\hat{\sigma}_z, \hat{\sigma}_\pm] = 2\hat{\sigma}_\pm \tag{56}$$

The unbolded $\hat{d}$ represents the component of the dipole moment operator $\hat{\boldsymbol{d}}$ and knowing that only the off-diagonal elements of the dipole operator are nonzero, since by parity consideration i.e. $\langle e|\hat{d}|e\rangle = \langle g|\hat{d}|g\rangle = 0$ where we define it as follows,

$$\begin{aligned}\hat{d} &= d_{ge}|g\rangle\langle e| + d_{eg}|e\rangle\langle g| \\ &= d\hat{\sigma}_- + d^*\hat{\sigma}_+ = d(\hat{\sigma}_+ + \hat{\sigma}_-)\end{aligned} \tag{57}$$

where $d_{ge} = d_{eg}^* = d$ (complex conjugate) since $\hat{\boldsymbol{d}}$ is Hermitian.

d_{ge} and d_{eg} are basically the matrix elements of dipole moment, $\langle g|\hat{d}|e\rangle$ and $d_{eg} = \langle e|\hat{d}|g\rangle$ and without the loss of generality to assume d is real. Therefore, the interaction Hamiltonian, H_I is now given as,

$$H_I = \hbar\lambda(\hat{\sigma}_+ + \hat{\sigma}_-)(\hat{a} + \hat{a}^+) \tag{58}$$

where $\lambda = \frac{dg}{\hbar}$. The expansion of term $(\hat{\sigma}_+ + \hat{\sigma}_-)(\hat{a} + \hat{a}^+)$ will lead to rapidly and slowly oscillating transition. We need to eliminate the rapidly oscillating term which is known as Rotating Wave Approximation (RWA).

Rotating Wave Approximation (RWA).

RWA is closely related the two-level approximation, assuming that the frequency of the field is in near resonance with the atomic transition, $\omega_L \approx \omega_0$. The condition where RWA serves as a good approximation is when the radiation field has low intensity so that the coupling strength is very small i.e. $\omega_0 \gg g$. Say the Rabi frequency, $\Omega_R(t) = -(\hat{d}_{eg})\frac{\hat{E}}{\hbar}$ which proportional to g and the phase are slowly differing in time in the order of $1/\omega_L$ or larger. $\Omega_R(t)$ is the term modulating the pulse. The broad pulse in a temporal domain i.e. a small modulation of plane wave refers to a slowly varying Rabi frequency. Thus, we approximate the pulse to be a continuous wave or quasi-monochromatic (T.K Shuen, Master thesis 2013). Let us first expand the term in the interaction Hamiltonian that is,

$$(\hat{\sigma}_+ + \hat{\sigma}_-)(\hat{a} + \hat{a}^+) = \hat{\sigma}_+\hat{a} + \hat{\sigma}_+\hat{a}^+ + \hat{\sigma}_-\hat{a} + \hat{\sigma}_-\hat{a}^+ \tag{59}$$

In the free-field case, the operators above evolve as,

$$\hat{a}(t) = \hat{a}(0)e^{-i\omega_L t} \;\; ; \;\; \hat{a}^+(t) = \hat{a}^+(0)e^{+i\omega_L t}$$

Same goes for the free-atomic case,

$$\hat{\sigma}_\pm(t) = \hat{\sigma}_\pm(0)e^{\pm i\omega_0 t}$$

Hence, we could see the approximation of time dependence of the operator after expansion in eq. (59),

$$\hat{\sigma}_+\hat{a} = \hat{\sigma}_+(0)\hat{a}(0)e^{i(\omega_0-\omega_L)} \sim e^{i(\omega_0-\omega_L)} \tag{i}$$

$$\hat{\sigma}_+\hat{a}^+ = \hat{\sigma}_+(0)\hat{a}^+(0)e^{i(\omega_0+\omega_L)} \sim e^{i(\omega_0+\omega_L)} \qquad (ii)$$

$$\hat{\sigma}_-\hat{a} = \hat{\sigma}_-(0)\hat{a}(0)\, e^{-i(\omega_0+\omega_L)} \sim e^{-i(\omega_0+\omega_L)} \qquad (iii)$$

$$\hat{\sigma}_-\hat{a}^+ = \hat{\sigma}_-(0)\hat{a}^+(0)\, e^{-i(\omega_0-\omega_L)} \sim e^{-i(\omega_0-\omega_L)} \qquad (iv)$$

Realize that in near resonance, the term in (ii & iii) oscillate more rapidly than the other two. Moreover, these terms violate the conservation of energy i.e. for (ii) the atomic excitation leads to photon emission and for (iii) the atomic de-excitation contributes to the absorption of photon. Such terms are not really observable and non-physical (D. Burgarth et al., 2024) so we should omit these terms as follows.

$$\hat{\sigma}_+\hat{a} + \hat{\sigma}_-\hat{a}^+ \qquad (60)$$

Hence the complete Jaynes-Cummings model for two-level atom is,

$$H = \frac{1}{2}\hbar\omega_0\hat{\sigma}_z + \hbar\omega_L\hat{a}^+\hat{a} + \hbar\lambda(\hat{\sigma}_+\hat{a} + \hat{\sigma}_-\hat{a}^+\,) \qquad (61)$$

The interaction Hamiltonian, can only cause transition of type

$$|e\rangle|n\rangle \leftrightarrow |g\rangle|n+1\rangle \text{ or } |e\rangle|n-1\rangle \leftrightarrow |g\rangle|n\rangle$$

We are going to use this as our model of two-level atom to analyze the dissipative mechanisms such as spontaneous emission and cavity loss using Lindblad Master equation and try to compare the results when using the usual Schrodinger equation (with no dissipation). We now continue to find the time-evolution wavefunction of the atom-field system and density matrix. For this case, we need to solve the eigenvalue and eigenvector equation as such.

$$H\Psi = E\Psi \qquad (62)$$

It is convenient for us to first find the matrix form of H by calculating all four matrix elements. We suppose the product state, also known as bare state to be,

$$|\Psi_{en}\rangle = |e\rangle|n\rangle = |e,n\rangle$$
$$|\Psi_{gn}\rangle = |g\rangle|n+1\rangle = |g,n+1\rangle \qquad (63)$$

The product state is normalized and orthogonal such that $\langle\Psi_{in}|\Psi_{jn}\rangle = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$ where $i,j = e,g$.

We can use the basis to find the matrix elements of H such that $H_{ij}^n = \langle\Psi_{in}|H|\Psi_{jn}\rangle$ as demonstrated below.

$$H_{ee}^n = \langle\Psi_{en}|H|\Psi_{en}\rangle$$
$$= \langle e,n|\frac{1}{2}\hbar\omega_0\hat{\sigma}_z|e,n\rangle + \langle e,n|\hbar\omega_L\hat{a}^+\hat{a}|e,n\rangle + \langle e,n|\hbar\lambda(\hat{\sigma}_+\hat{a} + \hat{\sigma}_-\hat{a}^+\,)|e,n\rangle$$
$$H_{ee}^n = \hbar\left(n\omega_L + \frac{\omega_0}{2}\right) \qquad (64)$$

Similarly for the other elements, we will have,

$$H_{eg}^n = \hbar\lambda\sqrt{n+1}$$
$$H_{ge}^n = \hbar\lambda\sqrt{n+1}$$

$$H_{gg}^{n} = \hbar\left((n+1)\omega_L - \frac{\omega_0}{2}\right)$$

Hence the matrix form of Jaynes-Cummings Hamiltonian is,

$$H = \begin{pmatrix} H_{ee}^{n} & H_{eg}^{n} \\ H_{ge}^{n} & H_{gg}^{n} \end{pmatrix} = \hbar\begin{pmatrix} \left(n\omega_L + \frac{\omega_0}{2}\right) & \lambda\sqrt{n+1} \\ \lambda\sqrt{n+1} & \left((n+1)\omega_L - \frac{\omega_0}{2}\right) \end{pmatrix} \tag{65}$$

Now we can conveniently calculate for the eigenvalue (energy) and eigenfunction (wave function) of the atom-field system using eq. (65) using determinant method.

$$|H - IE|\Psi = 0 \tag{66}$$

where I is the identity matrix and E is the eigenenergy.

$$\left|\hbar\begin{pmatrix} \left(n\omega_L + \frac{\omega_0}{2}\right) & \lambda\sqrt{n+1} \\ \lambda\sqrt{n+1} & \left((n+1)\omega_L - \frac{\omega_0}{2}\right) \end{pmatrix} - \begin{pmatrix} E & 0 \\ 0 & E \end{pmatrix}\right| = 0$$

$$= \left|\hbar\begin{pmatrix} \left(n\omega_L + \frac{\omega_0}{2}\right) - E & \lambda\sqrt{n+1} \\ \lambda\sqrt{n+1} & \left((n+1)\omega_L - \frac{\omega_0}{2}\right) - E \end{pmatrix}\right| = 0$$

$$= \hbar^2\left[\left(n\omega_L + \frac{\omega_0}{2}\right) - E\right]\left[\left((n+1)\omega_L - \frac{\omega_0}{2}\right) - E\right] - \left(\lambda\sqrt{n+1}\right)^2 = 0 \tag{67}$$

Simplifying to get,

$$E^2 - \hbar\omega_L(2n+1)E + \hbar^2\left\{n(n+1)\omega^2 - n\omega_L\omega_0 - \frac{\omega_0^2}{4} - \lambda^2(n+1)\right\} = 0 \tag{68}$$

Indeed, this is quadratic equation with the solution of,

$$E_{\pm} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{69}$$

where $a = 1,\ b = \hbar\omega_L(2n+1), c = \hbar^2\left\{n(n+1)\omega^2 - n\omega_L\omega_0 - \frac{\omega_0^2}{4} - \lambda^2(n+1)\right\}$

Substituting this into quadratic formula to get the eigenenergies,

$$E_{\pm} = \hbar\omega_L\left(n + \frac{1}{2}\right) \pm \frac{\hbar}{2}\Omega_{\mathrm{n}} \tag{70}$$

where $\Omega_{\mathrm{n}} = \sqrt{\Delta^2 + 4\lambda^2(n+1)}$ is the generalized Rabi frequency which depends on photon number n and $\Delta = |\omega_0 - \omega_L|$ is the detuning.

Now, the eigenstate is the dressed state of the atom-field system, which is the superposition of the bare states,

$$|+,n\rangle = \alpha_+|\Psi_{en}\rangle + \beta_+|\Psi_{gn}\rangle$$
$$|-,n\rangle = \alpha_-|\Psi_{en}\rangle + \beta_-|\Psi_{gn}\rangle \tag{71}$$

The dressed states above are obtained by using the orthonormality conditions that is,

$$|\alpha_+|^2 + |\beta_+|^2 = 1 \ ; \ \ |\alpha_-|^2 + |\beta_-|^2 = 1$$

For a sinusoidal solution, we notice that $\alpha_+ = \beta_- = \cos(\theta_n)$ and $\beta_+ = -\alpha_- = \sin(\theta_n)$ to be working out. Our task it to find the mixing angle which is θ_n.

$$|+,n\rangle = \cos\,(\theta_n)|\Psi_{en}\rangle + \sin\,(\theta_n)|\Psi_{gn}\rangle$$
$$|-,n\rangle = -\sin\,(\theta_n)|\Psi_{en}\rangle + \cos\,(\theta_n)|\Psi_{gn}\rangle \tag{72}$$

The mixing angle, θ_n is obtained to be,

$$\theta_n = \frac{1}{2}\tan^{-1}\left(\frac{2g\sqrt{n+1}}{\Delta}\right) = \frac{1}{2}\tan^{-1}\left(\frac{\Omega_n(0)}{\Delta}\right) \tag{73}$$

The trick to obtain θ_n is by simply solving the eigen-value and eigen-vector problem such that,

$$\hbar\begin{pmatrix}\left(n\omega_L + \frac{\omega_0}{2}\right) & \lambda\sqrt{n+1} \\ g\sqrt{n+1} & \left((n+1)\omega_L - \frac{\omega_0}{2}\right)\end{pmatrix}\begin{pmatrix}\alpha_\pm \\ \beta_\pm\end{pmatrix} = E_\pm\begin{pmatrix}\alpha_\pm \\ \beta_\pm\end{pmatrix} \tag{74}$$

such that $\alpha_\pm + \beta_\pm = 1$(Orthonormality condition).

Table 4.1 : Important identities

Quantity	Definition
Creation operators	$a^+\,\|n\rangle = \sqrt{n+1}\|n+1\rangle$
Annihilation operators	$a\,\|n\rangle = \sqrt{n}\|n-1\rangle$
Orthonormality condition	$\langle\Psi_i\|\Psi_j\rangle = \delta_{ij}$
Coherent state	$\|\alpha\rangle = \sum_{n=0} \alpha^n \frac{e^{-\frac{\|\alpha\|^2}{2}}}{\sqrt{n!}}\|n\rangle$
Density operator	$\rho = \|\Psi\rangle\langle\Psi\|$
Mean photon number	$\langle\hat{n}\rangle = \bar{n} = \langle\alpha\|\hat{n}\|\alpha\rangle = \|\alpha\|^2$
Atomic raising operators	$\sigma_+ = \|e\rangle\langle g\| = \begin{pmatrix}0 & 0 \\ 1 & 0\end{pmatrix}$
Atomic lowering operators	$\sigma_- = \|g\rangle\langle e\| = \begin{pmatrix}0 & 1 \\ 0 & 0\end{pmatrix}$

The table above shows important quantum mechanics identities used in this report in which some are cited from (F. H. A. Mathkoor, 2020, PhD thesis).

We can also derive the time-evolution wavefunction of this atom-field system in Fock state basis.

$$|\Psi_{\mathrm{af}}(0)\rangle = \sum_{m,n} |\Psi_{\mathrm{f}}(0)\rangle \otimes |\Psi_{\mathrm{a}}\rangle \qquad (75)$$

where $\Psi_{\mathrm{af}}(0)$ is atom-field wavefunction at t = 0, $\Psi_{\mathrm{f}}(0)$ and $\Psi_{\mathrm{a}}(0)$ are the wavefunction of atom and field respectively at t = 0.

Since the H_{int} couples only the states $|b, n+1\rangle$ and $|a, n\rangle$, we try the following ansatz, according to M.O Scully's and M.S.Zubairy's book,

$$|\Psi_{\mathrm{af}}(t)\rangle = \sum_{n=0} \left[C_g(t)C_n(t)|g,n\rangle + C_e(t)C_n(t)|e,n\rangle\right] \qquad (76)$$

Observe that the system is prepared such that the atom in excited state has one more photon then when in ground state. For simplicity, we let,

$$C_g(t)C_n(t) = C_{gn}(t);\ C_e(t)C_n(t) = C_{en}(t) \qquad (77)$$

Using time-dependent Schrodinger's equation, considering only the interaction Hamiltonian which contains all the dynamics of the system, one can get the coefficient to be in matrix form,

$$i\hbar \begin{pmatrix} \dot{C}_{g,n+1}(t) \\ \dot{C}_{en}(t) \end{pmatrix} = \hbar \begin{pmatrix} 0 & \lambda\sqrt{n+1}e^{-i(\omega_0-\omega_L)t} \\ \lambda\sqrt{n+1}e^{i(\omega_0-\omega_L)t} & 0 \end{pmatrix} \begin{pmatrix} C_{gm} \\ C_{en} \end{pmatrix}$$

$$\begin{cases} \dot{C}_{g,n+1}(t) = -i\lambda\sqrt{n+1}e^{-i(\omega_0-\omega_L)t}C_{en}(t) \\ \dot{C}_{en}(t) = -i\lambda\sqrt{n+1}e^{i(\omega_0-\omega_L)t}C_{g,n+1}(t) \end{cases} \qquad (78)$$

We must solve the coupled ODEs above simultaneously using Laplace Transform method where we can automatically include the initial conditions i.e. $C_{g,n+1}(0)$ and $C_{en}(0)$.

$$\bar{C}_{g,n+1}(s) = \int_0^\infty e^{-st}e^{-i(\omega_0-\omega_L)t}\, C_{gm}(t)\, dt$$

$$\bar{C}_{en}(s) = \int_0^\infty e^{-st}e^{i(\omega_0-\omega_L)t}\, C_{en}(t)\, dt \qquad (79)$$

The integral above will give us in the Laplace domain s, so we need to inverse Laplace Transform to get the solution in time domain, t. We do this using Python programming which turns out to be,

$$\begin{cases} C_{g,n+1}(t) = e^{-\frac{i\Delta}{2}t}\left\{C_{g,n+1}(0)\left[\cos\left(\frac{\Omega_{\mathrm{n}}}{2}t\right) + \frac{i\Delta}{\Omega_n}\sin\left(\frac{\Omega_{\mathrm{n}}}{2}t\right)\right] - iC_{en}(0)\frac{2\lambda\sqrt{n+1}}{\Omega_{\mathrm{n}}}\sin\left(\frac{\Omega_{\mathrm{n}}}{2}t\right)\right\} \\ C_{en}(t) = e^{\frac{i\Delta}{2}t}\left\{C_{en}(0)\left[\cos\left(\frac{\Omega_{\mathrm{n}}}{2}t\right) - \frac{i\Delta}{\Omega_n}\sin\left(\frac{\Omega_{\mathrm{n}}}{2}t\right)\right] - iC_{g,n+1}(0)\frac{2\lambda\sqrt{n+1}}{\Omega_{\mathrm{n}}}\sin\left(\frac{\Omega_{\mathrm{n}}}{2}t\right)\right\} \end{cases} \qquad (80)$$

In preparing the atom in excited state initially so the initial conditions demand that $C_{en}(0) = C_n(0)$ and $C_{g,n+1}\,(0) = 0$. $C_n(0)$ means that the probability amplitude of the field alone. We can adjust this to be any photon statistic we desire such as coherent state. Therefore, applying the initial conditions we obtain,

$$\begin{cases} C_{g,n+1}(t) = e^{-\frac{i\Delta}{2}t}\left\{-iC_n(0)\dfrac{2\lambda\sqrt{n+1}}{\Omega_{\mathrm{n}}}\sin\left(\dfrac{\Omega_{\mathrm{n}}}{2}t\right)\right\} \\ C_{en}(t) = e^{\frac{i\Delta}{2}t}\left\{C_n(0)\left[\cos\left(\dfrac{\Omega_{\mathrm{n}}}{2}t\right) - \dfrac{i\Delta}{\Omega_n}\sin\left(\dfrac{\Omega_{\mathrm{n}}}{2}t\right)\right]\right\} \end{cases} \tag{81}$$

We can easily obtained for the $\left|C_{g,n+1}(t)\right|^2, |C_{en}(t)|^2, C_{en}(t)C^*_{g,n+1}(t)$ and $C_{g,n+1}(t)C^*_{en}(t)$ as shown,

$$\left|C_{g,n+1}(t)\right|^2 = p_{nn}\frac{4\lambda^2(n+1)}{\Omega_n^2}\sin^2\left(\frac{\Omega_n}{2}t\right)$$

$$|C_{en}(t)|^2 = p_{nn}(\cos^2\left(\frac{\Omega_n}{2}t\right) + \left(\frac{\Delta}{\Omega_{\mathrm{n}}}\right)^2\sin^2\left(\frac{\Omega_{\mathrm{n}}}{2}t\right)$$

$$C_{en}(t)C^*_{g,n+1}(t) = ie^{i\Delta t}\frac{2\lambda\sqrt{n+1}}{\Omega_n}p_{nn}\left[\frac{1}{2}\sin(\Omega_n t) - \frac{i\Delta}{\Omega_n}\sin^2\left(\frac{\Omega_n}{2}t\right)\right]$$

$$C_{g,n+1}(t)C^*_{en}(t) = -ie^{-i\Delta t}\frac{2\lambda\sqrt{n+1}}{\Omega_n}p_{nn}\left[\frac{1}{2}\sin(\Omega_n t) + \frac{i\Delta}{\Omega_n}\sin^2\left(\frac{\Omega_n}{2}t\right)\right] \tag{82}$$

where $p_{nn} = C_n(0)$ depending on photon statistic used. Then we shift the (n+1) to n for the case of ground state as such.

$$\left|C_{gn}(t)\right|^2 = p_{n-1,n-1}\frac{4\lambda^2 n}{\Omega_{n-1}^2}\sin^2\left(\frac{\Omega_{n-1}}{2}t\right)$$

$$C_{en}(t)C^*_{gn}(t) = ie^{i\Delta t}\frac{2\lambda\sqrt{n}}{\Omega_{n-1}}p_{n,n-1}\left[\frac{1}{2}\sin(\Omega_n t) - \frac{i\Delta}{\Omega_n}\sin^2\left(\frac{\Omega_n}{2}t\right)\right]$$

$$C_{gn}(t)C^*_{en}(t) = -ie^{-i\Delta t}\frac{2\lambda\sqrt{n}}{\Omega_{n-1}}p_{n-1,n}\left[\frac{1}{2}\sin(\Omega_n t) + \frac{i\Delta}{\Omega_n}\sin^2\left(\frac{\Omega_n}{2}t\right)\right] \tag{83}$$

where $\Omega_{n-1} = \sqrt{\Delta^2 + 4\lambda^2 n}$ for the shifted n value generalized Rabi frequency.

For coherent state case, it is given by,

$$|\alpha\rangle = \sum_{n=0}\alpha^n\frac{e^{-\frac{|\alpha|^2}{2}}}{\sqrt{n!}}|n\rangle \tag{84}$$

The probability density for coherent state case is,

$$p_{nn} = |\langle n|\alpha\rangle|^2 = \frac{|\alpha|^{2n}e^{-|\alpha^2|}}{n!} = \frac{\langle n\rangle^n e^{-\langle n\rangle}}{n!}$$

$$p_{n-1,n-1} = |\langle n-1|\alpha\rangle|^2 = \frac{|\alpha|^{2(n-1)}e^{-|\alpha^2|}}{(n-1)!} = \frac{\langle n\rangle^{(n-1)}e^{-\langle n\rangle}}{(n-1)!}$$

$$p_{n,n-1} = p_{n-1,n} = \sqrt{\frac{\langle n\rangle^{2n-1}e^{-2\langle n\rangle}}{n!\,(n-1)!}} \tag{85}$$

where $\langle n\rangle = |\alpha|^2$ is the average photon number and α is the probability amplitude for coherent state.

Hence the density operator for the atom-field system is given by,

$$\begin{pmatrix} \rho_{ee} & \rho_{eg} \\ \rho_{ge} & \rho_{gg} \end{pmatrix} = \begin{pmatrix} |C_{en}(t)|^2 & C_{en}(t)C_{gn}^*(t) \\ C_{gn}(t)C_{en}^*(t) & \left|C_{gn}(t)\right|^2 \end{pmatrix} \tag{86}$$

We can check that the wavefunction in eq.(76) is normalized such that $\langle\Psi_{\mathrm{af}}(t)|\Psi_{\mathrm{af}}(t)\rangle = 1$ where

$$\rho_{gg} = \left|C_{gn}(t)\right|^2 (probability\ of\ being\ in\ ground\ state\ with\ n\ photon)$$

$$\rho_{ge}(t) = \rho_{eg}^*(t) = C_{gn}(t)C_{en}^*(t)\ (coherence)$$

$$\rho_{ee} = |C_{en}(t)|^2 (probability\ of\ being\ in\ excited\ state\ with\ n\ photon)$$

Using the wavefunction obtained we get,

$$\rho(t) = \begin{pmatrix} p_{nn}(\cos^2\left(\frac{\Omega_n}{2}t\right) + \left(\frac{\Delta}{\Omega_{\mathrm{n}}}\right)^2 \sin^2\left(\frac{\Omega_{\mathrm{n}}}{2}t\right) & ie^{i\Delta t}\frac{2\lambda\sqrt{n}}{\Omega_{n-1}}p_{n,n-1}\left[\frac{1}{2}\sin(\Omega_n t) - \frac{i\Delta}{\Omega_n}\sin^2\left(\frac{\Omega_n}{2}t\right)\right] \\ -ie^{-i\Delta t}\frac{2\lambda\sqrt{n}}{\Omega_{n-1}}p_{n-1,n}\left[\frac{1}{2}\sin(\Omega_n t) + \frac{i\Delta}{\Omega_n}\sin^2\left(\frac{\Omega_n}{2}t\right)\right] & p_{n-1,n-1}\frac{4\lambda^2 n}{\Omega_{n-1}^2}\sin^2\left(\frac{\Omega_{n-1}}{2}t\right) \end{pmatrix} \tag{87}$$

Checking to make sure that $\rho(t) = 1$ and $\rho^2(t) = 1$ (A pure state case)

For spontaneous emission case, we take a vacuum n = 1 and $p_{nm} = 1$.

$$\rho(t) = \begin{pmatrix} (\cos^2\left(\frac{\Omega_0}{2}t\right) + \left(\frac{\Delta}{\Omega_0}\right)^2 \sin^2\left(\frac{\Omega_0}{2}t\right) & ie^{i\Delta t}\frac{2\lambda\sqrt{n}}{\Omega_{n-1}}\left[\frac{1}{2}\sin(\Omega_n t) - \frac{i\Delta}{\Omega_n}\sin^2\left(\frac{\Omega_n}{2}t\right)\right] \\ -ie^{-i\Delta t}\frac{2\lambda\sqrt{n}}{\Omega_{n-1}}\left[\frac{1}{2}\sin(\Omega_n t) + \frac{i\Delta}{\Omega_n}\sin^2\left(\frac{\Omega_n}{2}t\right)\right] & \frac{4\lambda^2 n}{\Omega_{n-1}^2}\sin^2\left(\frac{\Omega_{n-1}}{2}t\right) \end{pmatrix} \tag{88}$$

Density operator of field from the density operator, $\boldsymbol{\rho(t)}$.

From the Von Neumann equation, the solution of $\rho(t)$ in the form below,

$$\rho(t) = U(t)\rho(0)U^+(t) \tag{89}$$

The atom-field density matrix at any time t is given by the expression,

$$\rho(t) = \rho_F(t) \otimes \rho_A(t) \tag{90}$$

where $\rho_F(t), \rho_A(t)$ is the density matrix of the field and atom respectively. Using the Von Neumann equation, we get the matrix element of $\rho(t)$ to be,

$$\begin{pmatrix} \dot{\rho_{ee}}(t) & \dot{\rho_{eg}}(t) \\ \dot{\rho_{ge}}(t) & \dot{\rho_{gg}}(t) \end{pmatrix} = -i/\hbar \left[\begin{pmatrix} \rho_{ee}(t) & \rho_{eg}(t) \\ \rho_{ge}(t) & \rho_{gg}(t) \end{pmatrix} \begin{pmatrix} \hbar\left(n\omega_L + \frac{\omega_0}{2}\right) & \hbar\lambda\sqrt{n+1} \\ \hbar\lambda\sqrt{n+1} & \hbar\left((n+1)\omega_L - \frac{\omega_0}{2}\right) \end{pmatrix} - \begin{pmatrix} \hbar\left(n\omega_L + \frac{\omega_0}{2}\right) & \hbar\lambda\sqrt{n+1} \\ \hbar\lambda\sqrt{n+1} & \hbar\left((n+1)\omega_L - \frac{\omega_0}{2}\right) \end{pmatrix} \begin{pmatrix} \rho_{ee}(t) & \rho_{eg}(t) \\ \rho_{ge}(t) & \rho_{gg}(t) \end{pmatrix} \right] \tag{91}$$

$$= \begin{pmatrix} i\lambda\left(\rho_{ge}(t) - \rho_{eg}(t)\right)\sqrt{n+1} & i\lambda\sqrt{n+1}\left(\rho_{gg}(t) - \rho_{ee}(t)\right) + i\rho_{eg}(\omega_0 - \omega_L) \\ i\lambda\sqrt{n+1}\left(\rho_{ee}(t) - \rho_{gg}(t)\right) + i\rho_{ge}(\omega_L - \omega_0) & i\lambda\left(\rho_{eg}(t) - \rho_{ge}(t)\right)\sqrt{n+1} \end{pmatrix}$$

Hence, we have the following coupled ODEs to solve,

$$\begin{aligned} \dot{\rho_{ee}}(t) &= i\lambda\big(\rho_{ge}(t) - \rho_{eg}(t)\big)\sqrt{n+1} \\ \dot{\rho_{eg}}(t) &= i\lambda\sqrt{n+1}\big(\rho_{gg}(t) - \rho_{ee}(t)\big) + i\rho_{eg}(\omega_0 - \omega_L) \\ \dot{\rho_{ge}}(t) &= i\lambda\sqrt{n+1}\big(\rho_{ee}(t) - \rho_{gg}(t)\big) + i\rho_{ge}(\omega_L - \omega_0) \\ \dot{\rho_{gg}}(t) &= i\lambda\left(\rho_{eg}(t) - \rho_{ge}(t)\right)\sqrt{n+1} \end{aligned} \tag{92}$$

We can easily solve the four coupled ODEs to obtain the respective density matrices using Laplace Transform from python code. We obtained the results as such,

$$\begin{aligned} \rho_{ee}(t) &= \frac{\lambda^2 + (\lambda^2 e^{-t\Omega_0} + 2\lambda^2 + \Delta^2)e^{-t\Omega_0}}{\Omega_0^2} \cdot e^{t\Omega_0} \\ \rho_{eg}(t) &= \frac{\lambda(-\omega_0 e^{-2t\Omega_0} + 2\omega_0 e^{-t\Omega_0} - \omega_0 + \omega_L e^{-2t\Omega_0})}{2\Omega_0^2} - 2\omega_L e^{-t\Omega_0} + \omega_L - i\Omega_0(e^{-2t\Omega_0} - e^{t\Omega_0}) \\ \rho_{ge}(t) &= \rho_{eg}^*(t) \\ \rho_{gg}(t) &= \frac{\lambda^2(-e^{-2t\Omega_0} + 2e^{-t\Omega_0} - 1)e^{t\Omega_0}}{\Omega_0^2} \end{aligned} \tag{93}$$

$$\rho_{ee}(t) + \rho_{gg}(t) = 1\,;\rho(t) > 0\,;\rho_{eg} = \rho_{ge}^*$$

where $\Omega_0 = \sqrt{\Delta^2 + 4\lambda^2}$ is the generalized vacuum Rabi frequency.

Then the atomic inversion $w(t) = \rho_{ee}(t) - \rho_{gg}(t)$

The density operator with dissipation :

To obtain the density operator of the atom-field system considering the dissipation effect due spontaneous emission and cavity decay we should use the Master equation in Lindblad form as shown,

$$\frac{d}{dt}\rho(t) = -\frac{i}{\hbar}[H,\rho(t)] + \mathcal{L}_{\text{sp}}[\rho] + \mathcal{L}_{\text{cav}}[\rho] \tag{94}$$

where

$$\mathcal{L}_{\text{sp}}[\rho(t)] = \gamma\left[\sigma_-\rho(t)\sigma_+ - \frac{1}{2}\{\sigma_+\sigma_-,\rho(t)\}\right] \qquad (94a)$$

$$\mathcal{L}_{\text{cav}}[\rho(t)] = \kappa\left[a\rho(t)a^+ - \frac{1}{2}\{a^+a,\rho(t)\}\right] \qquad (94b)$$

are Liouvillian operators for spontaneous emission and cavity loss respectively where γ and κ the rate of dissipation. Firstly, we consider the master equation for spontaneous emission first that is eq. (94a). The tedious derivation of Lindblad master equation is derived in appendix B1. We can derive master equation for spontaneous emission as follows using matrix method.

$$\begin{pmatrix} \dot{\rho_{ee}}(t) & \dot{\rho_{eg}}(t) \\ \dot{\rho_{ge}}(t) & \dot{\rho_{gg}}(t) \end{pmatrix} = -\frac{i}{\hbar}[H,\rho(t)]$$
$$+\gamma\left[\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} \rho_{ee}(t) & \rho_{eg}(t) \\ \rho_{ge}(t) & \rho_{gg}(t) \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \frac{1}{2}\left(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} \rho_{ee}(t) & \rho_{eg}(t) \\ \rho_{ge}(t) & \rho_{gg}(t) \end{pmatrix} + \begin{pmatrix} \rho_{ee}(t) & \rho_{eg}(t) \\ \rho_{ge}(t) & \rho_{gg}(t) \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\right)\right]$$
$$= \begin{pmatrix} i\lambda\big(\rho_{ge}(t) - \rho_{eg}(t)\big)\sqrt{n+1} - \rho_{ee}(t)\gamma & i\lambda\sqrt{n+1}\big(\rho_{gg}(t) - \rho_{ee}(t)\big) + i\rho_{eg}(\omega_0 - \omega_L) - \frac{1}{2}\rho_{eg}(t)\gamma \\ i\lambda\sqrt{n+1}\big(\rho_{ee}(t) - \rho_{gg}(t)\big) + i\rho_{ge}(\omega_L - \omega_0) - \frac{1}{2}\rho_{ge}(t)\gamma & i\lambda\big(\rho_{eg}(t) - \rho_{ge}(t)\big)\sqrt{n+1} + \rho_{ee}\gamma \end{pmatrix} \tag{95}$$

which leads to the following coupled ODEs below.

$$\dot{\rho_{ee}}(t) = i\lambda\big(\rho_{ge}(t) - \rho_{eg}(t)\big)\sqrt{n+1} - \rho_{ee}(t)\gamma$$
$$\dot{\rho_{eg}}(t) = i\lambda\sqrt{n+1}\big(\rho_{gg}(t) - \rho_{ee}(t)\big) + i\rho_{eg}(\omega_0 - \omega_L) - \frac{1}{2}\rho_{eg}(t)\gamma$$
$$\dot{\rho_{ge}}(t) = i\lambda\sqrt{n+1}\big(\rho_{ee}(t) - \rho_{gg}(t)\big) + i\rho_{ge}(\omega_L - \omega_0) - \frac{1}{2}\rho_{ge}(t)\gamma$$
$$\dot{\rho_{gg}}(t) = i\lambda\left(\rho_{eg}(t) - \rho_{ge}(t)\right)\sqrt{n+1} + \rho_{ee}\gamma \tag{96}$$

Next for the cavity decay, the master equation is given by,

$$\frac{d}{dt}\rho(t) = -\frac{i}{\hbar}[H,\rho(t)] + \kappa\left[a\rho(t)a^+ - \frac{1}{2}\{a^+a,\rho(t)\}\right] \tag{97}$$

The general solution of this master equation according to (S. M. Barnett & J. Jeffers, 2010) is given by,

$$\rho(t) = \sum_{m=0}^{\infty} \kappa^m \int_0^t dt_m \int_0^{t_m} dt_{m-1} \dots \int_0^{t_2} dt_1\, e^{-i\hat{\mathcal{H}}(t-t_m)}\hat{a}$$
$$\times\, e^{-i\hat{\mathcal{H}}(t_m-t_{m-1})}\hat{a} \dots \hat{a} e^{-i\hat{\mathcal{H}}t_1}\rho(0)e^{i\hat{\mathcal{H}}t_1}\hat{a}^\dagger \dots \hat{a}^\dagger e^{i\hat{\mathcal{H}}(t_m-t_{m-1})}\hat{a}^\dagger e^{i\hat{\mathcal{H}}(t-t_m)} \tag{98}$$

For this research, we are going to take the matrix element from the master equation. This is quite a trivial procedure where we act on the bra and ket as provided in appendix **B2** :

$$\rho_{ij}^{mn}(t) = \langle i, m|\rho(t)|j, n\rangle \tag{99}$$

where i,j is the atomic states and m,n is photonic state. In general, the Lindblad Master Equation for cavity damping is given by,

$$\dot{\rho}_{gg}^{mn}(t) = -i\left[(m-n)\omega_L\dot{\rho}_{gg}^{mn}(t) + \lambda\left(\sqrt{m}\dot{\rho}_{eg}^{m-1,n}(t) - \sqrt{n}\dot{\rho}_{ge}^{m,n-1}(t)\right)\right]$$
$$+\,\kappa\left(\dot{\rho}_{gg}^{m+1,n+1}(t)\sqrt{(m+1)(n+1)} - \frac{(m+n)}{2}\dot{\rho}_{gg}^{mn}(t)\right)$$
$$\dot{\rho}_{ge}^{mn}(t) = -i\left[(-\omega_0 + (m-n)\omega_L)\dot{\rho}_{gg}^{mn}(t) + \lambda\left(\sqrt{m}\dot{\rho}_{ee}^{m-1,n}(t) - \sqrt{n+1}\dot{\rho}_{gg}^{m,n+1}(t)\right)\right]$$
$$+\,\kappa\left(\dot{\rho}_{ge}^{m+1,n+1}(t)\sqrt{(m+1)(n+1)} - \frac{(m+n)}{2}\dot{\rho}_{ge}^{mn}(t)\right)$$
$$\dot{\rho}_{eg}^{mn}(t) = -i\left[(\omega_0 + (m-n)\omega_L)\dot{\rho}_{gg}^{mn}(t) + \lambda\left(\sqrt{m+1}\dot{\rho}_{gg}^{m+1,n}(t) - \sqrt{n}\dot{\rho}_{ee}^{m,n-1}(t)\right)\right]$$
$$+\,\kappa\left(\dot{\rho}_{eg}^{m+1,n+1}(t)\sqrt{(m+1)(n+1)} - \frac{(m+n)}{2}\dot{\rho}_{eg}^{mn}(t)\right)$$
$$\dot{\rho}_{ee}^{mn}(t) = -i\left[(m-n)\omega_L\dot{\rho}_{ee}^{mn}(t) + \lambda\left(\sqrt{m+1}\dot{\rho}_{ge}^{m+1,n}(t) - \sqrt{n+1}\dot{\rho}_{eg}^{m,n+1}(t)\right)\right]$$
$$+\,\kappa\left(\dot{\rho}_{ee}^{m+1,n+1}(t)\sqrt{(m+1)(n+1)} - \frac{(m+n)}{2}\dot{\rho}_{ee}^{mn}(t)\right) \tag{100}$$

The difficulty faced here where we cannot straightforwardly solve the coupled ODEs since we have the matrix element with difference photon number indices i.e. (m+1), (n+1), (m-1) and (n-1). As stated by Watanabe, A., & Nakazato, H. (2022), usually the quantum harmonic oscillator represented by a Fock space where the a and a^+ operators are represented as infinite dimensional matrices.Thus, for this reason, solving Lindblad master equation is extremely difficult in Hilbert space. Due to this infinite dimensionality, one has to impose a truncation of the Hilbert space. So, we now try to shift the index for $\dot{\rho}_{ge}^{mn}(t)$, $\dot{\rho}_{eg}^{mn}(t)$ and $\dot{\rho}_{ge}^{mn}(t)$ and get a close form quickly.

$$\dot{\rho}_{ge}^{m,n-1}(t) = -i\left[(-\omega_0 + (m-n+1)\omega_L)\dot{\rho}_{gg}^{m,n-1}(t) + \lambda\left(\sqrt{m}\dot{\rho}_{ee}^{m-1,n-1}(t) - \sqrt{n}\dot{\rho}_{gg}^{m,n}(t)\right)\right]$$
$$+\kappa\left(\dot{\rho}_{ge}^{m+1,n}(t)\sqrt{(m+1)(n)} - \frac{(m+n-1)}{2}\dot{\rho}_{ge}^{m,n-1}(t)\right)$$
$$\dot{\rho}_{eg}^{m-1,n}(t) = -i\left[(\omega_0 + (m-n-1)\omega_L)\dot{\rho}_{gg}^{m-1,n}(t) + \lambda\left(\sqrt{m}\dot{\rho}_{gg}^{m,n}(t) - \sqrt{n}\dot{\rho}_{ee}^{m-1,n-1}(t)\right)\right]$$
$$+\kappa\left(\dot{\rho}_{eg}^{m,n+1}(t)\sqrt{(m)(n+1)} - \frac{(m+n-1)}{2}\dot{\rho}_{eg}^{m-1,n}(t)\right)$$
$$\dot{\rho}_{ee}^{m-1,n-1}(t) = -i\left[(m-n)\omega_L\dot{\rho}_{ee}^{m-1,n-1}(t) + \lambda\left(\sqrt{m}\dot{\rho}_{ge}^{m,n-1}(t) - \sqrt{n}\dot{\rho}_{eg}^{m-1,n}(t)\right)\right]$$
$$+\kappa\left(\dot{\rho}_{ee}^{m,n}(t)\sqrt{(m)(n)} - \frac{(m+n-2)}{2}\dot{\rho}_{ee}^{m-1,n-1}(t)\right)$$

(101)

To see this clearly, we can represent in matrix form as follows,

$$\begin{pmatrix} \dot{\rho}_{ee}^{m-1,n-1}(t) \\ \dot{\rho}_{eg}^{m-1,n}(t) \\ \dot{\rho}_{ge}^{m,n-1}(t) \\ \dot{\rho}_{gg}^{mn}(t) \end{pmatrix}$$
$$= \begin{pmatrix} -i(m-n)\omega_L & i\lambda\sqrt{n} & -i\lambda\sqrt{m} & 0 \\ i\lambda\sqrt{n} & -i(\omega_0 + (m-n-1)\omega_L) - \kappa\left(\frac{m+n-1}{2}\right) & 0 & -i\lambda\sqrt{m} \\ -i\lambda\sqrt{m} & 0 & -i(\omega_0 + (m-n+1)\omega_L) - \kappa\left(\frac{m+n-1}{2}\right) & i\lambda\sqrt{n} \\ 0 & -i\lambda\sqrt{m} & i\lambda\sqrt{n} & -i(m-n)\omega_L - \frac{\kappa(m-n)}{2} \end{pmatrix} \begin{pmatrix} \rho_{ee}^{m-1,n-1}(t) \\ \rho_{eg}^{m-1,n}(t) \\ \rho_{ge}^{m,n-1}(t) \\ \rho_{gg}^{m,n}(t) \end{pmatrix}$$
$$+\kappa\begin{pmatrix} \sqrt{mn}\rho_{ee}^{m,n}(t) \\ \sqrt{(m)(n+1)}\rho_{eg}^{m,n+1}(t) \\ \sqrt{(m+1)(n)}\rho_{ge}^{m+1,n}(t) \\ \sqrt{(m+1)(n+1)}\rho_{gg}^{m+1,n+1}(t) \end{pmatrix}$$

(102)

where we let the $\rho_{ee}^{m,n}(t)$, $\rho_{eg}^{m,n+1}(t)$, $\rho_{ge}^{m+1,n}(t)$ and $\rho_{gg}^{m+1,n+1}(t)$ to be the solution for the atom-field system without dissipation effect i.e. taken from the Schrodinger equation method. This is the approach we figured out during the research in order to solve the unclosed form of the density matrix element due to infinite recursion of Hilbert space. However, this method provides an approximation only for the density matrix elements, so there will be a significant error at some extent though. Now we can proceed to solve the coupled ODEs to get the solution for $\rho_{ee}^{m-1,n-1}(t)$, $\rho_{eg}^{m-1,n}(t)$, $\rho_{ge}^{m,n-1}(t)$ and $\rho_{gg}^{m,n}(t)$.

4.3 The Dynamic of Two-level atom

We can explore interesting phenomena in conjunction with investigation of decoherence mechanism such as the collapse and revival phenomena. Most of the textbook discussed these phenomena without the presence of damping. Therefore, the atomic inversion is given by W(t) for the case of without dissipation as per below,

$$W(t) = \langle \Psi(t)|\widehat{\sigma_3}|\Psi(t)\rangle = P_e(t) - P_g(t)$$
$$= \left\{ p_{nn}\left(\cos^2\left(\frac{\Omega_n}{2}t\right) + \left(\frac{\Delta}{\Omega_\mathrm{n}}\right)^2 \sin^2\left(\frac{\Omega_\mathrm{n}}{2}t\right)\right)\right\} - p_{n-1,n-1}\frac{4\lambda^2 n}{\Omega_{n-1}^2}\sin^2\left(\frac{\Omega_{n-1}}{2}t\right) \tag{103}$$

But $\Omega_\mathrm{n}^2 = (\omega_0 - \omega_L)^2 + 4g^2(n+1)$ and driven by coherent state p_{nn}. For a vacuum state, $p_{nn} = 1$. With dissipation, the results may differ. In general, we can implement the following formula into program to obtain the plot quickly,

$$W(t) = p_{nn}(\rho_{ee} - \rho_{gg}) \tag{104}$$

for spontaneous emission case and,

$$W(t) = p_{nn}\left(\rho_{ee}^{m-1,n-1}(t) - \rho_{ee}^{m,n}(t)\right) \tag{105}$$

for cavity decay loss. We weighed it at coherent state as our initial photon statistic.

The collapse and revival time can be easily computed, according to (C. C. Garry & P. L. Knight, 2005) is given by,

$$t_c = \frac{\sqrt{2}}{\lambda}\sqrt{\frac{\bar{n}+1}{\bar{n}}} \approx \frac{\sqrt{2}}{\lambda}\ ; \bar{n} \gg 1 \tag{106}$$

for the collapse time where $\bar{n}$ is mean photon number and for revival time is,

$$t_r = \left(\frac{2\pi}{\lambda}\right)(\bar{n}+1)^{\frac{1}{2}} \approx \left(\frac{2\pi}{\lambda}\right)(\bar{n})^{\frac{1}{2}}\ ; \bar{n} \gg 1 \tag{107}$$

The probability distribution has its peak at n = $\bar{n}(1 - \Gamma t)$. So in the presence of dissipation such as spontaneous emission, the modified t_c and t_r according to (S. M. Barnett & J. Jeffers, 2010),

$$t_c^\Gamma = \frac{\sqrt{2}}{\lambda}\sqrt{\frac{\bar{n}(1-\Gamma \mathrm{t_c^\Gamma}) + 1}{\bar{n}(1-\Gamma \mathrm{t_c^\Gamma})}}$$

$$t_r^\Gamma = t_r - \frac{2\pi^2 k^2 \bar{n}}{\lambda^2}\Gamma \tag{108}$$

where Γ is the rate of dissipation and k is wave number.

CHAPTER 5: RESULTS AND DISCUSSIONS

For this chapter, the results from the investigation of dynamics of Jaynes-Cummings Model are discussed in depth here. This chapter will unravel the phenomena such as avoided crossing, dissipation which causing decoherence in the system as well as collapse and revival with the consideration of dissipation or non-dissipation case.

5.1 Avoided Crossing

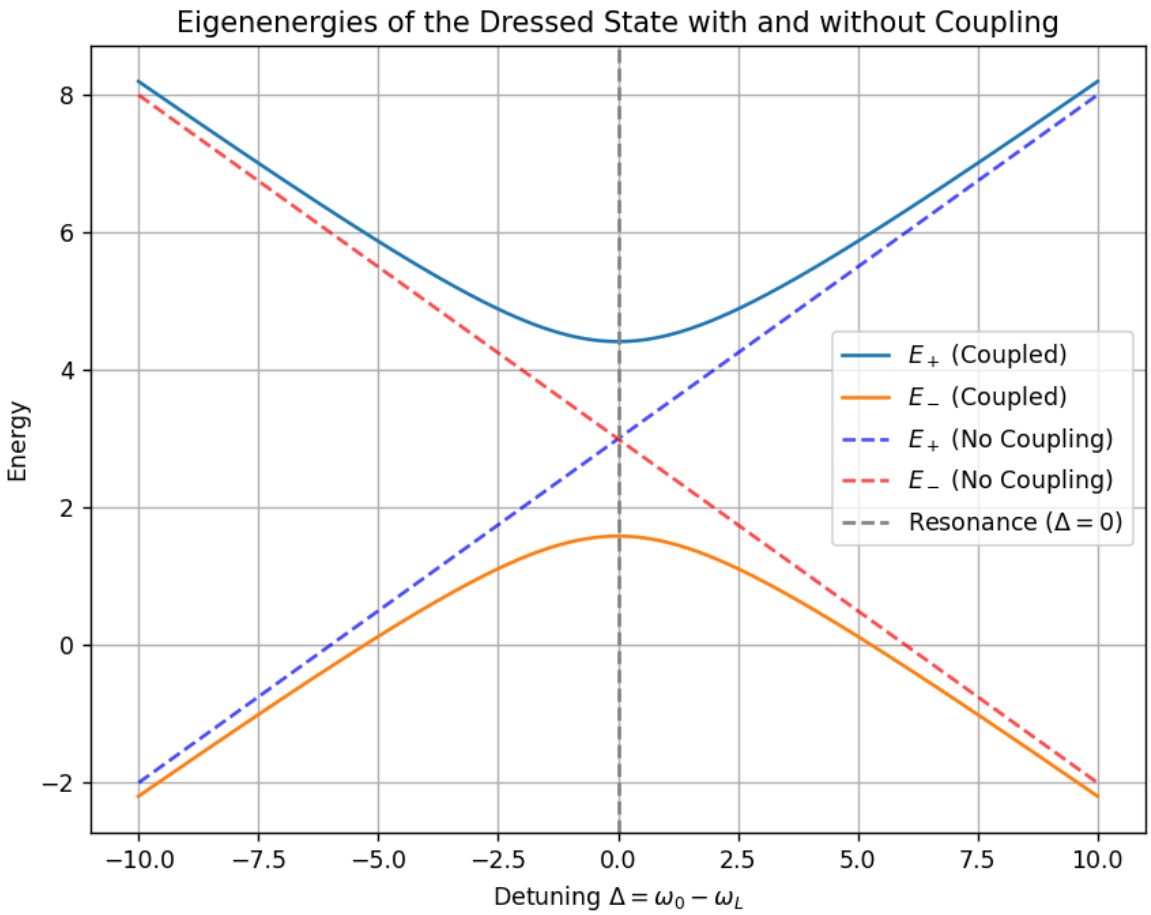


Figure 5.1 : The plot of eigenvalue of the dressed state from equation (70) which results in the quantum phenomenon called an avoided crossing.

Referring to figure 5.1, there are two types of lines i.e. the dash lines (blue and red) and the curved solid line (blue and orange). As we can observe, the dash lines will meet at a certain point i.e. at centre while the solid curve will never touch each other. This is called the avoided crossing phenomenon when the eigenenergies of the two coupled quantum system in which the atom-field system. The interaction between atom and field system will cause the energy level to repeal each other instead of intersecting. As shown in eq. (70), the energy eigenstate which corresponds to the two interacting atom-field system hybridized which then leads to what is known as ‘dressed state’. Dressed state is the quantum superposition of the ‘bare state’ or uncoupled states.

Referring to eq. (70), when we have the case of no coupling i.e. $\lambda = \frac{dg}{\hbar} = 0$ (the coupling strength), the eigenenergies of the system are simply just $E_{\pm} = \hbar\omega_L\left(n + \frac{1}{2}\right) \pm \frac{\hbar}{2}\Delta$ which is the solution for the straight line and intersect at $\Delta = 0$ i.e. the detuning is zero between the atomic transition frequency (ω_0) and the field frequency (ω_L) which is represented by the dashed lines in the graph. When we introduce coupling i.e. $\lambda \neq 0$, the energies of the dressed state split by an amount of proportional to the coupling strength, leading to the avoided crossing behaviour as observed in solid curved. The minimum separation at $\Delta = 0$ is calculated to be $2\lambda\hbar(n+1)^{1/2}$, choosing n = 0 (in vacuum state) and natural unit of $\hbar = 1$ we get the separation to be typically 2λ.

Moreover, at higher detuning Δ, the dash line seems like act like the asymptote of the solid curve. In other words, the dressed or couple state approaches or behaves like a uncoupled or bare state. This specifically occurs when $\Delta \gg \lambda$. At large detuning, the atom-field interaction becomes negligible so the avoided crossing is typically observed at zero detuning (resonant) or lower detuning (near-resonant). Hence the system behaves as if like an independent component rather than a strongly coupled hybrid.

5.2 Von Neumann Density matrix element plot (No Dissipation)

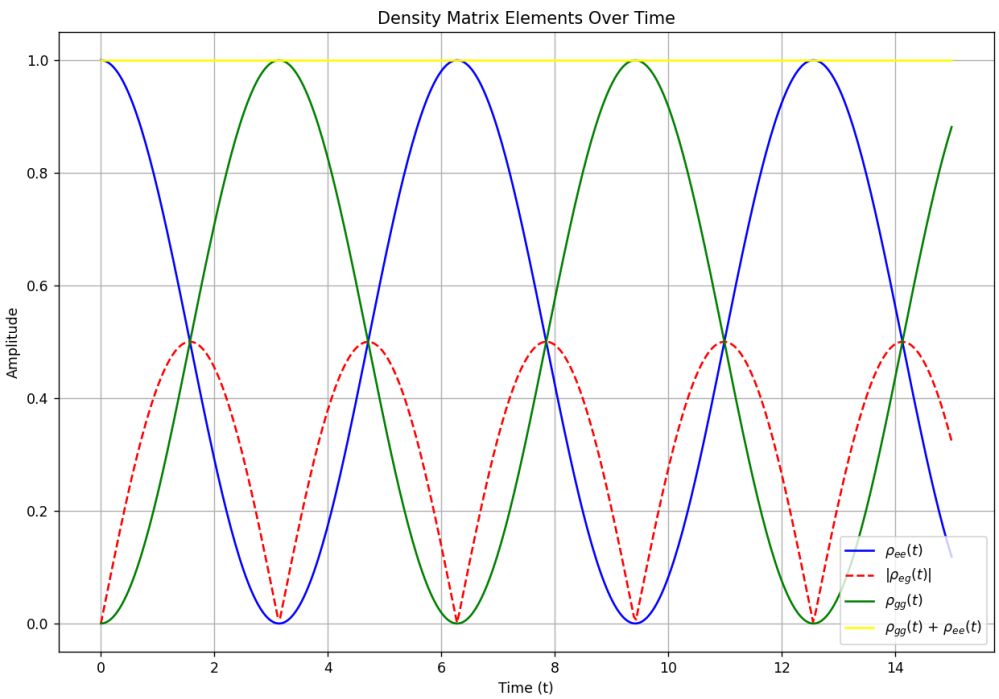


Figure 5.2 : The plot of the density matrix element representing the Rabi oscillation derived from the Von Neumann equation of the two-level atom interacting with EM field in the cavity with parameters $\omega_0 = \omega_L = 2, \lambda = 0.5, n = 0$.

The graph from Figure 5.2 was plotted using the Von Neumann equation from eq. (92). The similar plot can be produced as we used the density matrix form using Schrodinger equation from eq. (88) considering the vacuum state i.e. n = 0 and $C_n(0) = p_{nn} = 1$. The Von Neumann equation describes

how does the density operator evolves over time which will tell us much about the dynamics of the system.

The population plot from the density matrix element is represented by the blue and green colour solid lines. The blue one represents the excited state population ρ_{ee} while the green one represents the ground state population ρ_{gg}. This is not considering the effect of dissipation yet or still in an ideal condition. One thing to note that we prepared the system to be initially at excited state. So the initial condition implemented in the program is $\rho_{ee}(0) = 1$ and $\rho_{gg}(0) = 0$. At any time, our populations density matrix elements must add up to 1, i.e. $\rho_{ee}(t) + \rho_{gg}(t) = 1$ for our results to be valid. This characteristic is represented by the solid yellow curve above. The population elements basically reflect the periodic exchange of energy between the atom and the field with the atom transitioned between excited |e> and ground |g> states. The ground state $\rho_{gg}(t)$ oscillates out with respect to the excited state $\rho_{ee}(t)$ whereby becomes maximum when $\rho_{ee}(t)$ is at minimum and vice versa. The phenomenon shows the Rabi oscillation which is the oscillatory energy exchange between these population states. This occurs due to the coherent interaction between the atom and the field governed by Rabi frequency $\Omega_n = \sqrt{\Delta^2 + 4\lambda^2(n+1)}$. We can deduce the oscillation period or Rabi period as $T_n = \frac{1}{\sqrt{\Delta^2+4\lambda^2(n+1)}}$. When at huge detuning condition i.e. $\Delta \gg 1$, oscillation amplitude decreases, showing an incomplete population transfer due to reduction in coupling efficiency. We can prove this by the following graph in Figure 5.3 below. As we can see the amplitude of population and coherence decreases significantly.

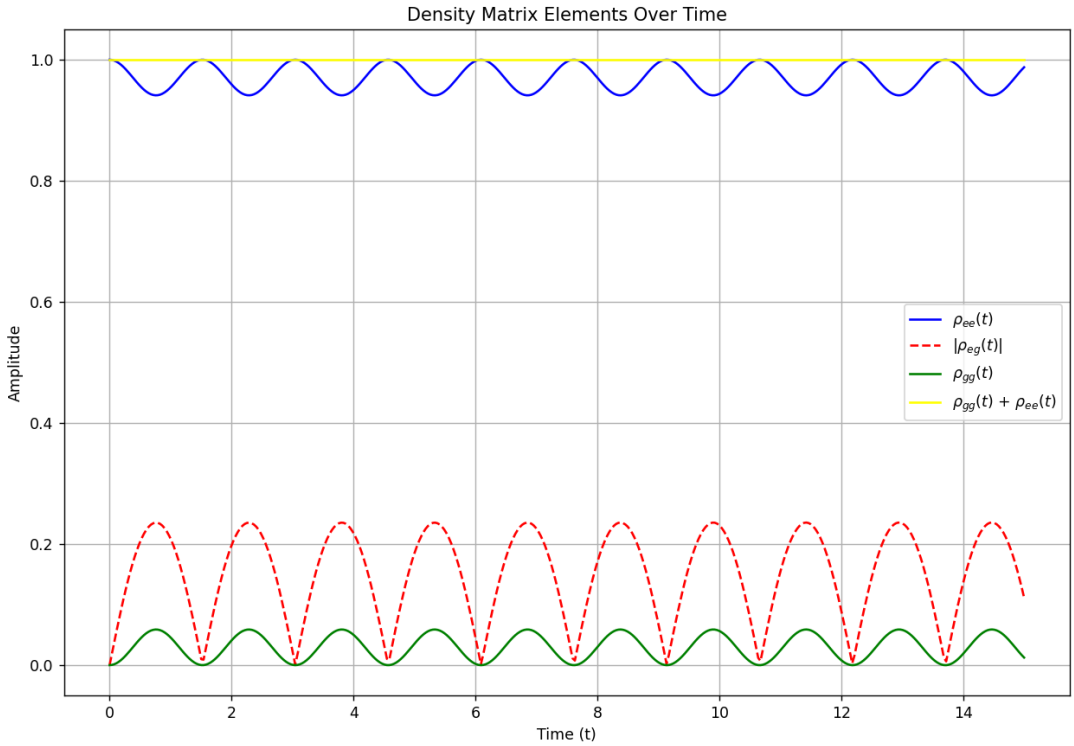


Figure 5.3 : The plot of the density matrix element representing the Rabi oscillation derived from the Von Neumann equation of the two-level atom interacting with EM field in the cavity with parameters $\omega_0 = 6, \omega_L = 2, \lambda = 0.5, n = 0$.

Next is the dash red line represents the coherence dynamics given by the magnitude of $|\rho_{eg}|$ and $|\rho_{ge}|$. Note that these matrix elements are imaginary that is in the form of a + ib, and they are conjugate with each other i.e. $\rho_{ge} = \rho_{eg}^*$ in Figure 5.2. Its peak is occurs at the point where the population are evenly distributed that is when $\rho_{ee} = \rho_{gg} = 0.5$ or when the system is in phase or full superposition. The dips occur when the atom is either fully in the ground state ($\rho_{gg} = 1$) or excited state ($\rho_{ee} = 1$). In short, the coherence element represents the quantum superposition of excited and ground state where its behaviour is tied to the degree of population mixing. Maximum coherence element would mean that a strong superposition while the minimum coherence element would mean it collapse into a pure state.

5.3 Density matrix plot (Spontaneous emission)

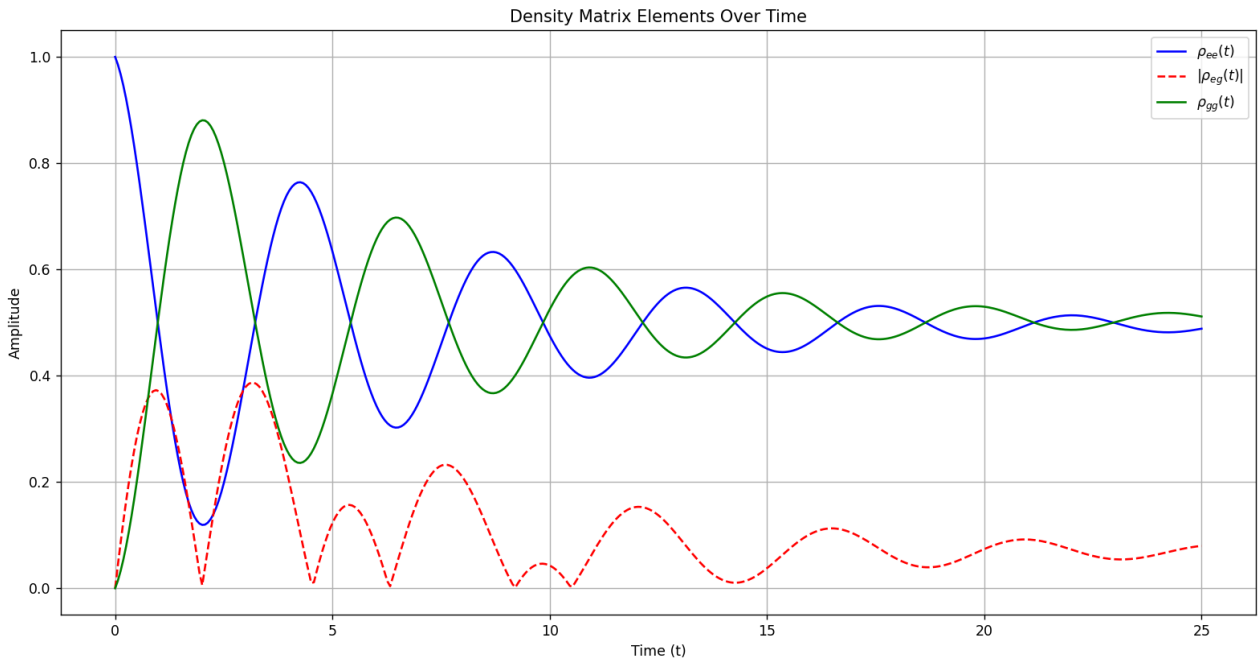


Figure 5.4 : The plot of the density matrix element representing the damped Rabi oscillation due to spontaneous emission derived from the Lindblad master equation of the two-level atom interacting with EM field in the cavity with parameters $\omega_0 = \omega_L = 2, \lambda = 0.5, \gamma = 0.1$ and $n = 0$.

The plot represents the evolution of density matrix elements for the two-level atom under the effect of spontaneous emission which is governed by Lindblad master equation in eq. (96). In Figure 5.2 and 5.3, we have seen the Rabi oscillation for non-dissipative case that is, $\gamma = 0$. We observed from Figure 5.4 the population and coherence of the density matrix elements damped over time due to spontaneous emission. Previously we have learnt in chapter 2 that spontaneous emission occurs due to the interaction of the quantum system (atom-field) with the vacuum state(n = 0). Initially the population of the excited state starts at maximum i.e. $\rho_{ee} = 1$ but then over time it decays exponentially indicated by the wave envelope due to spontaneous emission with oscillatory modulations caused by coherent energy exchange with field. At long time ($t \to \infty$), ρ_{ee} will stabilize and stop oscillating eventually indicating the atom relaxed to the ground state. The plot for $\rho_{ee}(t)$ is

represented by the solid blue curve whereas the solid green curve for $\rho_{gg}(t)$. Similarly, this phenomenon can also be observed for the case of ρ_{gg} too. The exponential decay due to spontaneous emission is actually governed by Weisskopf-Wigner theory. Note that, the sum of $\rho_{ee}(t) + \rho_{gg}(t)$ should always equal to 1, as obtained from the graph.

For the coherence element, $|\rho_{eg}(t)|$ represented by the solid red curve. The coherence also exhibit reduction in amplitude differs from when we have no dissipation at all depicted in Figure 5.2 in which the maximum amplitude is constant at 0.5. The oscillation in $\rho_{ee}(t), \rho_{gg}(t)\ and\ |\rho_{eg}(t)|$ signifies the dynamics of the Rabi oscillation modulated by the coupling strength $\lambda = 0.5$. The spontaneous emission rate $\gamma = 0.1$ reduces the oscillatory amplitude over time, resulting in a transition from coherent oscillatory behaviour to a steady state. The decay of $|\rho_{eg}(t)|$ on the other hand demonstrates the loss of quantum coherence due to spontaneous emission. The reduction of magnitude of coherence proving the decoherence in a damped JCM system.

Apart from that, as we increase the $\gamma = 1, 5, 10\ and\ 100$ as shown in Figure 5.5 (a)-(d) below. For instance, for small γ that is $\gamma < \lambda$ we can observe the density matrix elements plot still oscillate over time. But when $\gamma > \lambda$ (large damping), we can observe that the population of excited state undergoes critically damping in which the value approaches to 0. As the value of γ increases, the ρ_{ee} reduce till zero and the coherent superposition, indicated by the $|\rho_{eg}|$ is also destroyed.

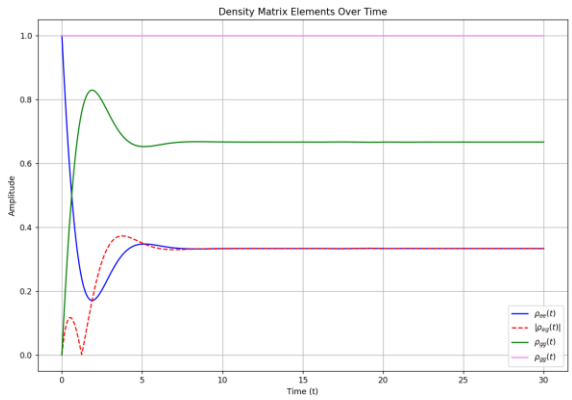


Figure 5. 5 (a) : $\gamma = 1$

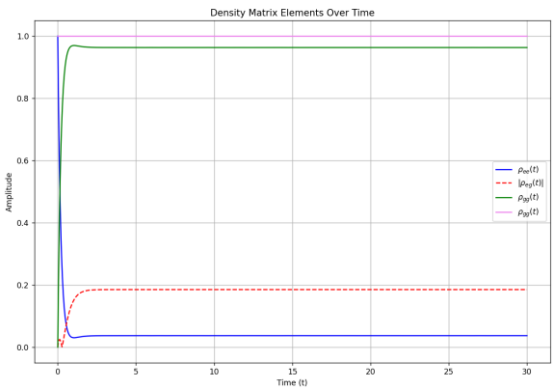


Figure 5.5 (b) : $\gamma = 5$

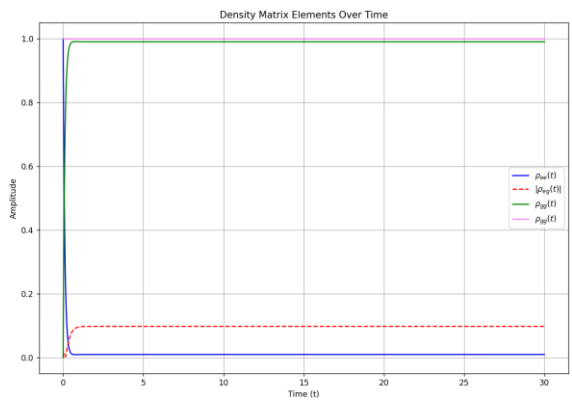


Figure 5.5 (c) : $\gamma = 10$

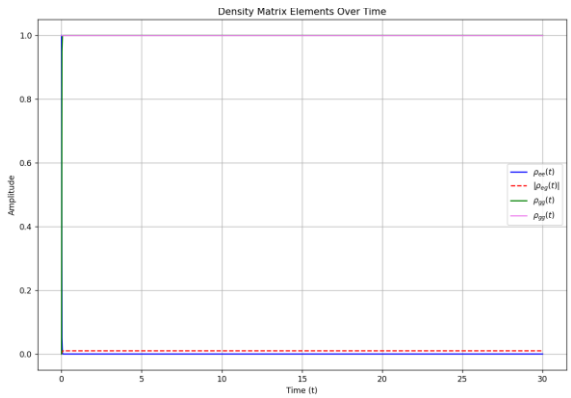


Figure 5.5 (d) : $\gamma = 100$

Figure 5.5 : The plot of the density matrix element representing the damped Rabi oscillation due to spontaneous emission with parameters $\omega_0 = \omega_L = 2, \lambda = 0.5, \gamma = 1, 5, 10, 100$ and $n = 0$.

5.4 Atomic inversion (no dissipation)

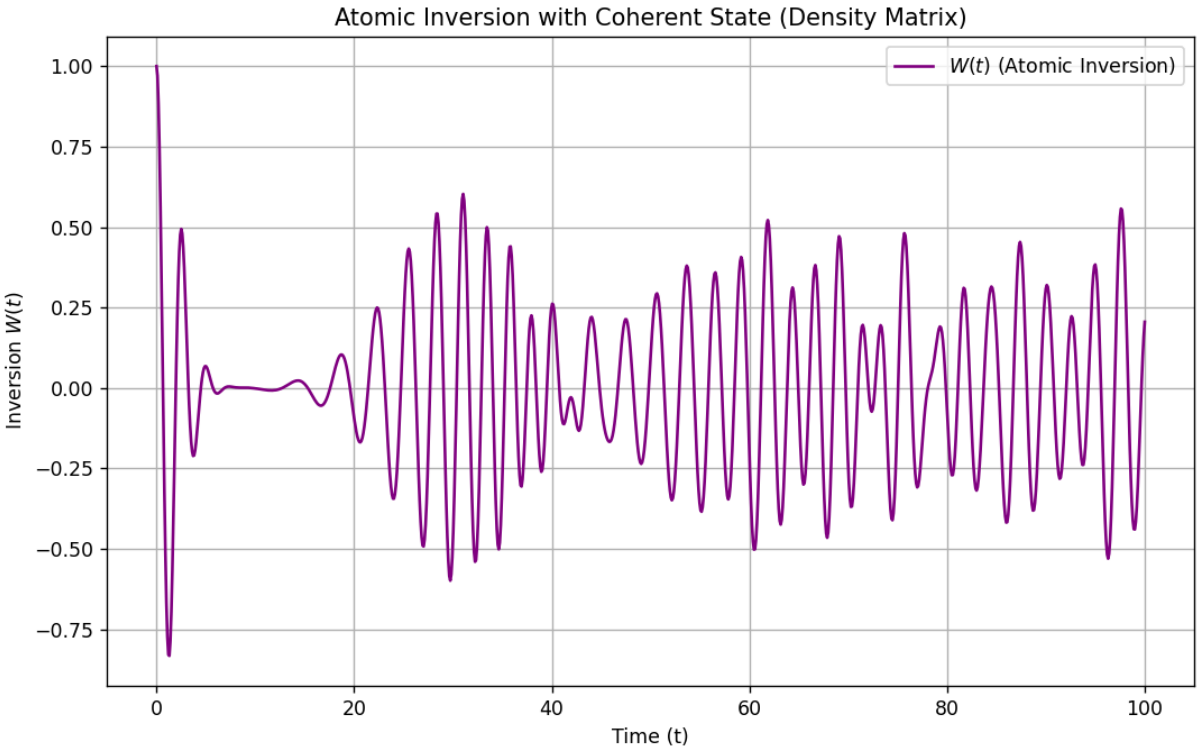


Figure 5.6 : Plot of atomic inversion W(t) with parameters $\omega_0 = \omega_L = 2, \lambda = 0.5, \gamma = 0$ and $n_{max} = 25$ and $\bar{n} =< n >= 5$.

5.5 Atomic Inversion (Spontaneous emission)

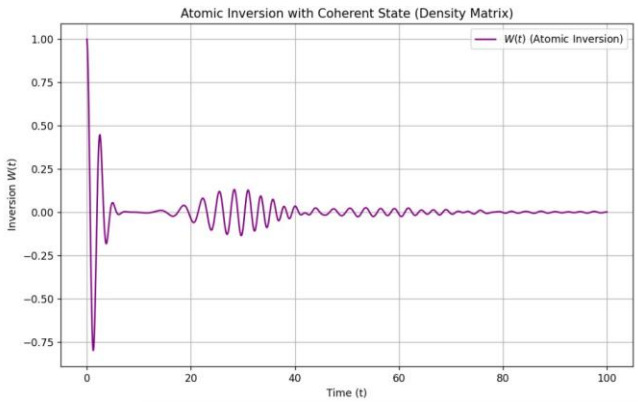


Figure 5.7 (a) : $\gamma = 0.1$

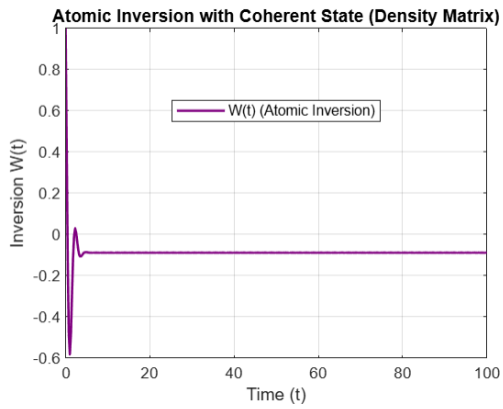


Figure 5.7 (b) : $\gamma = 1$

Figure 5.7 : Plot of atomic inversion W(t) with parameters $\omega_0 = \omega_L = 2, \lambda = 0.5$ and $n_{max} = 25$ and $\bar{n} =< n >= 5$.

5.6 Atomic Inversion (Spontaneous emission, <n> = 0)

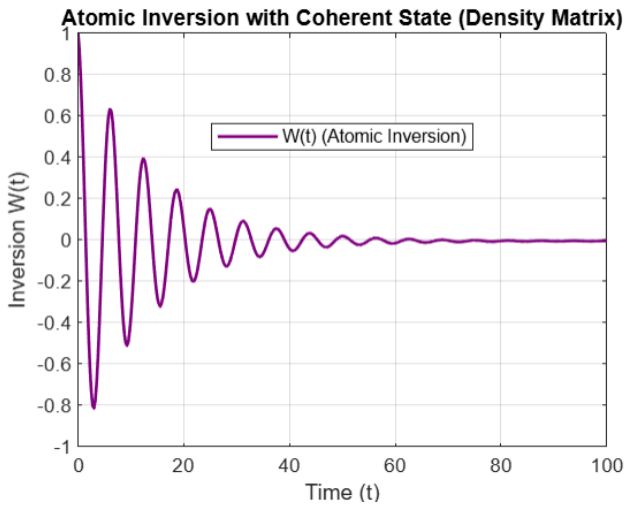


Figure 5.8 (a) : $\gamma = 0.1$

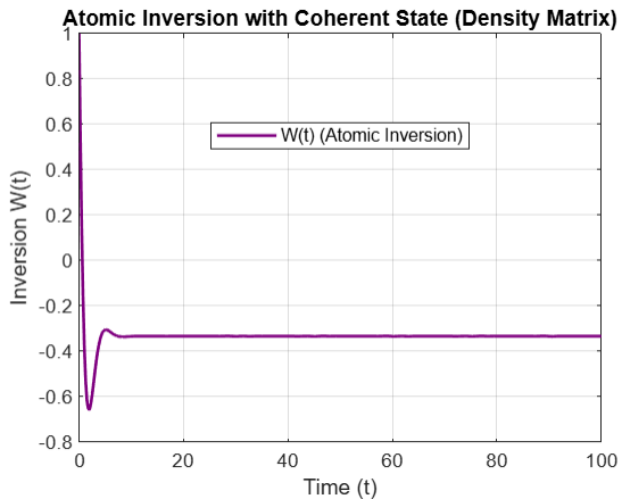


Figure 5.8 (b) : $\gamma = 1$

Figure 5.8 : Plot of atomic inversion W(t) in vacuum with parameters $\omega_0 = \omega_L = 2$, $\lambda = 0.5$ and $n_{max} = 0$ and $\bar{n} =< n >= 0$.

5.7 Atomic Inversion (cavity decay)

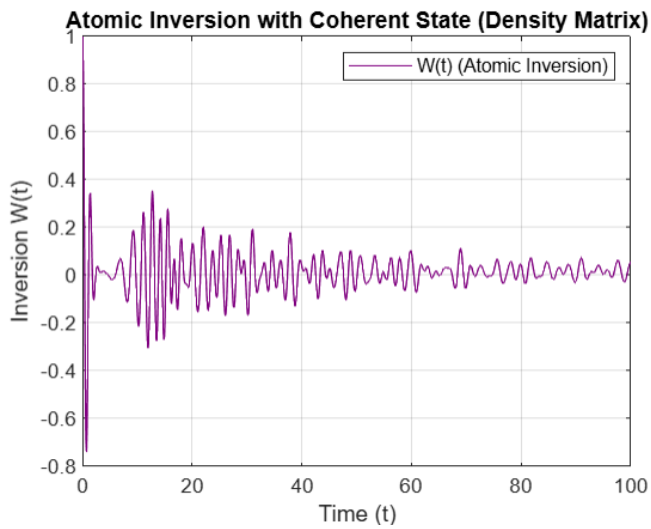


Figure 5.9 (a) : $\kappa = 0.01$

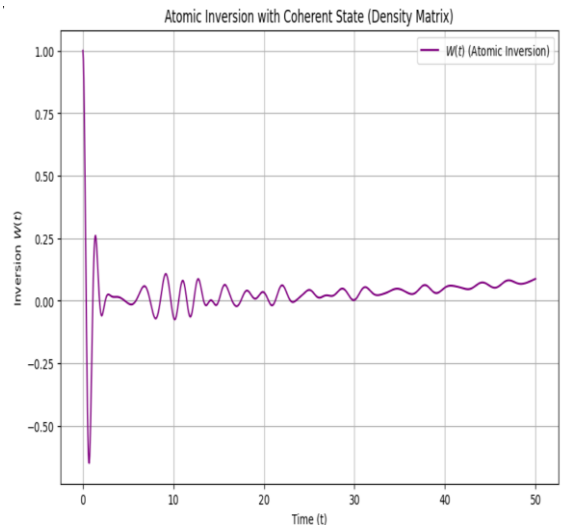


Figure 5.9 (b) : $\kappa = 0.05$

Figure 5. 9: Plot of atomic inversion W(t) for cavity decay with parameters $\omega_0 = \omega_L = 2$, and $n_{max} = 100$ and $\bar{n} =< n >= 5$.

In figure 5.6 – 5.9, we obtained the atomic inversion using eq. (104) which results in the quantum collapse and revival phenomenon. One of the most interesting phenomena can be collapse-revival effect in population inversion of atomic levels. It is given by the difference between excited state population and ground state population driven by coherent state or vacuum state ($when\ p_{nn} = 1$). In

the present case the envelope of the sinusoidal Rabi oscillations 'collapses' to zero. Nevertheless, over time we observed a 'revival' of the collapsed inversion. For Figure 5.6, we consider the atomic inversion with no dissipation with parameters $n_{max} = 25\ and\ \bar{n} =< n >= 5$. The phenomena of collapse and revival is repeated or oscillated over time whereby the amplitude of the Rabi oscillation reduces and time duration for the revival occurs increases then eventually overlaps with the previous revival. The revival and collapse time for non-dissipative case are given in the previous chapter. Moreover, this phenomenon can be simply understood from the formula eq. (104) where each terms in the summation (sigma over n) represents Rabi oscillation for integer of n. The p_{nn} is basically the photon distribution function in which determines the weight for each n value. Initially the atom is prepared in a definite states. So the n-terms summation are correlated. Over time, the Rabi oscillation associated with different excitations have different frequencies and hence become uncorrelated or out of phase causing collapse. Nevertheless, as time progresses, the correlation now restored back known as revival. The reason behind the occurrence of revivals is due to the 'granular' structure of photon distribution i.e. stream of photons. Hence the collapse and revival phenomena is a purely quantum phenomena in which no classical analogue.

For Figure 5.7(a) and (b), with the same parameters used except there exists spontaneous emission with $\gamma = 0.1$ and 1 respectively. As for the case of $\gamma = 0.1$ the collapse and revival still holds but with a reduction of amplitude. In this case, the collapse and revival occurs several times before the oscillation stops or dampens at 0 amplitude. The modified revival and collapse time for the dissipative case have been calculated in the previous chapter. For the case of $\gamma = 1$ instead, the Rabi oscillation does not even complete one oscillation to begin with before it completely stops at W(t) = -0.3. This indicates a strong damping factor as $\gamma > \lambda$. Overall, as the coherence matrix element undergoes decay, it leads to a reduction in the amplitude of the Rabi oscillations over time then ultimately driving the system to a steady-state value where inversion approaches zero. This behaviour is a direct result of spontaneous emission dissipating energy from the atom-field system into the environment. A similar result can be observed from the Figure 5.9 (a) and (b). For smaller $\kappa = 0.01$, we can see a collapse and revival characteristic once again but this time slightly smaller amplitude (of course) due to cavity decay caused by imperfect cavity mirror which allows photon to leak out and interact with external environment. But, eventually for both cases, the amplitude of the W(t) will be approaching zero as time progresses. The explanation for this is that when the system losing a photon, it will introduce a phase shift in Rabi oscillations and such that this shift depends on time at which quantum is lost. But, since we don't know the times when any quanta are lost, we need to add contribution from all possible time and hence this leads to suppress the resulting contribution to inversion.

Figure 5.8, on the other hand requires a separate explanation. The parameter used is $n_{max} = 0$, $\bar{n} = 0$(mean photon number) dictating we are considering for the case of vacuum state. Using the same formula of atomic inversion, this can be achieved by letting photon number to be zero and the coherent state distribution, p_{nn} to be initially 1. Hence this is said that the coherent state degenerates into vacuum state. The atomic inversion W(t) depicts a damped periodic oscillation at varying amplitude i.e. quantum-vacuum Rabi oscillation yet there is no collapse and revival phenomena observed compared from the previous case. For ideal case, we would expect the plot to be like an ideal oscillatory system to oscillate at constant maximum and minimum amplitude over time. For

vacuum case, this plot makes sense and logical since, as explained above that is each term in the summation associated with the Rabi oscillation for n photons with coherent state as the photon distribution. These photons has their own associated Rabi frequency and different frequency can cause uncorrelated (out of phase which leads to collapse) and becomes correlated back then revive. For vacuum state, there is no photon exists i.e. n = 0, so how can we expect to have a correlated or uncorrelated photons since we have no photon to begin with. But still, the Rabi oscillation undergoes damping as $\gamma = 0.1$ still smaller than λ. Nevertheless, as the spontaneous emission factor $\gamma = 1 > \lambda$, the system cannot oscillate and immediately dampen, it could not even undergo one oscillation.

5.8 Density matrix plot (cavity decay)

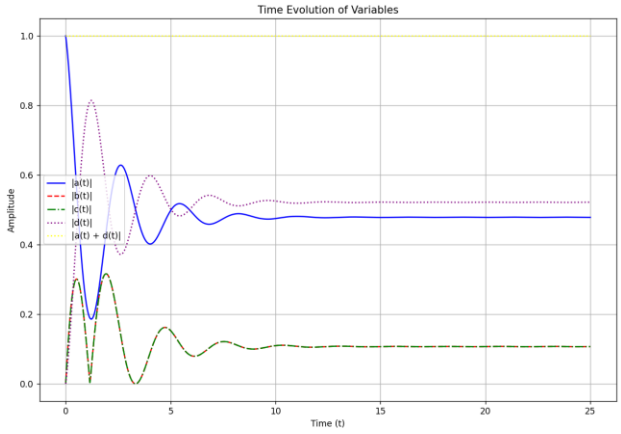


Figure 5.10 (a) : $\kappa = 0.1$

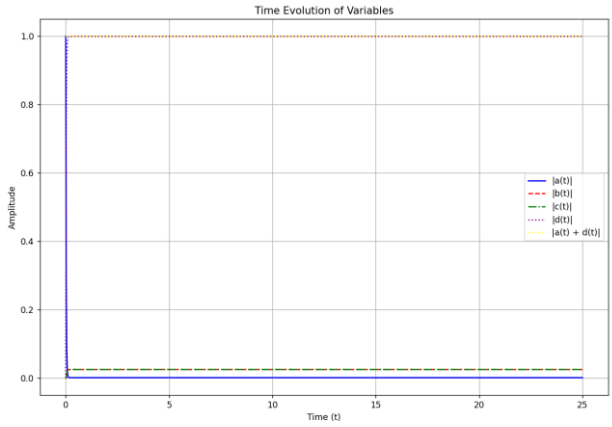


Figure 5.10 (b) : $\kappa = 10$

Figure 5.10 : The plot of the density matrix element representing the damped Rabi oscillation due to cavity decay with parameters $\omega_0 = \omega_L = 2, \lambda = 0.5, \gamma = 0.1, 10$ and $n = 5$.

We observed almost similar phenomenon for the case of evolution of density matrix over time as compared to dissipation due to spontaneous emission. For small cavity decay rate, κ the photon emitted by the atom stays for some time inside the optical cavity (might undergo multiple reflections from the mirror) and then it will escape at some moment. One might think the oscillation of ρ_{ee} and ρ_{gg} should decay to zero, but when we observe from the graph plotted above, the oscillation kept constant at some steady state value. This is not a wrong result obtained nor if it is due to computational error.

Instead, the nonzero steady-state amplitudes observed in both cases can be attributed to the nature of the quantum dissipative process. Cavity decay typically transfers energy from the cavity field to the environment however it does not completely depopulate the atomic states or fully suppress coherence. The results appear consistent with the expected behaviour of an atom-field system under cavity decay. A detailed check of the Lindblad terms and the total Hamiltonian used in the simulation would clarify whether the observed steady-state behaviour aligns with theoretical predictions. Moreover, as κ value very high i.e. $\kappa > \lambda$, the number of oscillations will be reduced, the photon will stays inside the optical cavity progressively for shorter time interval after its creation.

CHAPTER 6: MITIGATION TECHNIQUES OF DECOHERENCE

Spontaneous emission is an intrinsic property of matter that we can greatly have control on as it relies on the nature of the surrounding environment i.e. vacuum state (V. Weisskopf & E. Wigner, 1930). For the case of cavity decay due to photon absorption, scattering or loss, we can control the amount of decay by using different Q-factor or fidelity of the optical cavity. The effect of dissipation and decoherence is still an on-going challenge in quantum processing, communication and computation. Hence, in this chapter, we are going to explore several mitigation for dissipative mechanisms in quantum system.

6.1 Engineered photonic environment

Engineering the photonic environment is meant to modify the EM field or radiation surrounding of quantum emitters to control spontaneous emission. By embedding quantum emitters in photonic crystals or microcavities, the photonic density of states can be tailored to suppress or increase emission. The Purcell effect occurs when the local electromagnetic environment influences the spontaneous emission rate, γ is a crucial approach to mitigate spontaneous emission.

For instance, a photonic crystal with bandgap properties can block the propagation of light within specific frequency ranges, effectively reducing undesirable emission channels. Lodahl et al. has demonstrated this, where they achieved controlled emission dynamics using quantum dots embedded in photonic crystals. Similarly, we can also use a high-quality factor (high Q-factor) microcavities such as whispering-gallery mode resonators to provide enhanced confinement of light, resulting to a prolonged lifetime of photon and reduced the dissipation due to cavity decay (Vahala, 2003). Hence, these advancements are important for quantum systems where preserving coherence over extended periods is essential for reliable operation.

Apart from that, a nanofabrication technique has further refined these structures, making them scalable and practical for integration into complex quantum networks. As such, devices incorporating photonic crystals or microcavities have shown potential in securing quantum communication protocols and quantum memories, where maintaining coherence is a must.

6.2 Dynamical decoupling

Another way to mitigate decoherence effect by spontaneous emission and cavity decay is through dynamical decoupling (DD) technique. The technique mitigates decoherence by applying sequences of control pulses to a quantum system, effectively decoupling it from environmental noise. This approach compensates for system-environment interactions by averaging out noise effects over time.

Viola and Lloyd (1998) introduced the theoretical framework for this technique, and it has since been experimentally validated in various quantum systems. Another research by B. Pokharel et. al (2018), demonstrated the use of different DD type to suppress spontaneous emission and dephasing. They use the sequence of pulse $(ZI)^N$ to suppress spontaneous emission which resulting from an inevitable coupling of any quantum system to vacuum field.

Also, more advanced pulse sequences, such as concatenated DD and Uhrig DD, have been developed to address different noise spectra. For example, Uhrig's sequence optimizes pulse timings for systems dominated by low-frequency noise to enhance coherence time significantly. These methods have proven effective for platforms like trapped ions and superconducting qubits, which are prone to environmental disturbances.

The adaptability of DD to diverse noise models ensures its relevance across quantum technologies. It has been integrated into quantum processors and memories to improve error resilience. For example, trapped-ion systems equipped with DD protocols have achieved coherence times suitable for practical quantum computations (Häffner et al., 2008).

6.3 Utilization of strong coupling regimes

Besides that, we can also utilize a strong coupling region whereby in the strong coupling regime, the interaction strength between a quantum emitter and a cavity field surpasses the system's dissipation rates. This regime is characterized by coherent energy exchange between the emitter and cavity, leading to the formation of dressed states that reduce energy loss through spontaneous emission or photon leakage.

The Jaynes-Cummings model describes this regime's physics, which has been extensively studied in cavity quantum electrodynamics (QED) systems. Haroche and Raimond (2006) has demonstrated experimental realizations of strong coupling where they can achieve a controlled interactions that minimize decoherence or dissipation. Not only that, a strong coupling also enables phenomena such as vacuum Rabi splitting, which protects coherence by redistributing energy into stable dressed states.

The advancement in cavity design and atom-field control have made strong coupling regimes more accessible. This will lead to more broad applications such as quantum metrology and communication. By achieving robust energy exchange in these systems, quantum coherence can be preserved for longer durations, critical for high-precision quantum operations.

6.4 Environmental engineering

The final mitigation approach is by environmental engineering approach. This approach focuses on optimizing the surrounding environment to reduce cavity decay and spontaneous emission. High-Q cavities play a pivotal role by confining light with minimal photon leakage. These cavities, such as whispering-gallery mode resonators and photonic crystal cavities, are engineered to achieve exceptional light confinement.

The research by Vahala in 2003 highlighted how such structures significantly enhance photon lifetimes, allowing for improved coherence in quantum systems. Recent innovations in materials science, such as the use of low-loss dielectric materials, have further minimized energy dissipation in cavities. Additionally, tailoring boundary conditions and integrating these cavities into hybrid quantum systems have made them effective for scalable quantum devices.

Moreover, these engineered environments are particularly relevant for quantum networks, where photon loss during transmission can severely impact performance. High-Q cavities ensure that photons retain their coherence over long distances, making them indispensable for quantum communication and memory applications.

CHAPTER 7 : CONCLUSION

In conclusion, the dynamics of the Jaynes-Cummings model has been investigated and explored in this research in which focusing on the interplay between atom and cavity field under dissipation mechanisms such as the spontaneous emission and cavity decay. Also, we have shown the quantum phenomenon of dressed state (interaction between atom-field system) associated with the avoided crossing. In contrast with the uncoupled or bare state where the eigenvalues crossed at zero detuning. The crossing of the solid line in figure 5.1 indicating the degenerate energy at zero detuning between uncoupled atom and field.

Moreover, we also explore the evolution of density matrix first using Von Neumann equation for non-dissipative case and then master equation with additional dissipator term to obtain the evolution of density matrix element. By deriving the evolution of density matrix element, we can provide a insight into population redistribution, coherence decay and steady-state behaviour of atom-field system. We have seen that as the dissipation rate (γ, κ) increases the population density matrix elements will decay exponentially indicated by the envelope of the wave. For spontaneous emission, this fits according to Weisskopf-Wigner theory. The coherence matrix element $(\rho_{eg}\ , \rho_{ge})$ will also reduce over time and stop oscillating at some moment. When the dissipation rates become very large, this will cause destruction in the coherence term and the population of ρ_{ee} will undergo critically damping.

We also discovered the collapse and revival phenomena for both dissipative and non-dissipative case. In ideal case, the collapse and revival will keep progressing or repeating over time highlighting the coherent superposition nature of atom-field system. On the other hand, the presence of dissipation will suppress the collapse and revival and beside altering the population. For small γ and κ, the collapse and revival of the Rabi oscillation occurs several times but then eventually it dampens into 0. For higher dissipation rate, somehow the Rabi oscillation couldn't manage to revive and stop oscillating.

Lastly, we also discussed the mitigation strategy of decoherence and dissipation due to spontaneous emission and cavity decay. For instance, one can use a high-Q factor and high fidelity of optical cavity in order to control amount of photon loss. Not only that, we can also implement a typical method such as environmental engineering, utilizing a strong coupling regime, dynamical decoupling and engineered photonic environment.

REFERENCES

Marlan Orvil Scully, & Muhammad Suhail Zubairy. (2008). Quantum optics. Cambridge University Press

Gerry, C. C., & Knight, P. L. (2008). Introductory quantum optics. Cambridge University Press.

Cook, R. J., & Milonni, P. W. (1988). Spontaneous excitation of atoms near a phase conjugating mirror. IEEE Journal of Quantum Electronics, 24(7), 1383–1387. https://doi.org/10.1109/3.977

Nunn, Joshua. (2008). Quantum Memory in Atomic Ensemble. PhD thesis, Oxford university

Quantum Physics of Light-Matter Interactions Claudiu Genes. (n.d.). Retrieved January 12, 2025, from https://mpl.mpg.de/fileadmin/user_upload/LectureNotes.pdf

Heinz-Peter Breuer, & Petruccione, F. (2010). The theory of open quantum systems. Oxford Univ. Press.

Scala, M., Militello, B., Messina, A., Jyrki Piilo, & Maniscalco, S. (2007). Microscopic derivation of the JaynesCummings model with cavity losses. Physical Review A, 75(1). https://doi.org/10.1103/physreva.75.013811

T. K Shuen. (2013). Study of laser interaction with quantum particles. Master thesis, Universiti Malaya, Kuala Lumpur.

F. H. A. Mathkoor (2020). Quantum two-photon laser in kerr-like medium. PhD thesis, Universiti Malaya, Kuala Lumpur.

Watanabe, A., & Nakazato, H. (2022). Exact Master Equation for an Open Jaynes-Cummings System. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4043585

Lvovsky, A. I., Sanders, B. C., & Tittel, W. (2009). Optical quantum memory. *Nature Photonics*, *3*(12), 706–714. https://doi.org/10.1038/nphoton.2009.231

M. Schlosshauer (2019). Quantum Decoherence. University of Portland. https://doi.org/10.1016/j.physrep.2019.10.001

E. T. Jaynes and F. W. Cummings, “Comparison of quantum and semiclassical radiation theories with application to the beam maser,” Proc. IEEE 51, 89–109 (1963)

Mischuck, B., & Mølmer, K. (2013). Qudit quantum computation in the Jaynes-Cummings model. *Physical Review A*, *87*(2). https://doi.org/10.1103/physreva.87.022341

J. I. Cirac and P. Zoller, Phys. Rev. Lett. 74 (1995), 4091–4094.

Daniele De Bernardis, Mercurio, A., & Simone De Liberato. (2024). Nonperturbative cavity quantum electrodynamics: is the Jaynes-Cummings model still relevant? *Journal of the Optical Society of America B*. https://doi.org/10.1364/josab.522786

Chong, S. Y., & Shen, J. Q. (2019). Quantum collapse-revival effect in a supersymmetric Jaynes–Cummings model and its possible application in supersymmetric qubits. *Physica Scripta*, *95*(5), 055104–055104. https://doi.org/10.1088/1402-4896/ab5c6e

F. D. Bonani (2020). The Jaynes-Cummings Model. University of Sao Paulo, Brazil. Retrieved on https://ifsc.usp.br/~strontium/Teaching/Material2020-1%20SFI5814%20Atomicamolecular/Fabio%20-%20Monograph%20-%20Jaynes-Cummings%20model.pdf

Salatino, L., Mariani, L., Attanasio, C., Pagano, S., & Citro, R. (2023). Dissipative dynamics in quantum key distribution. *The European Physical Journal Plus*, *138*(6). https://doi.org/10.1140/epjp/s13360-023-04138-3

Eberly, J. H., Narozhny, N. B., & Sanchez-Mondragon, J. J. (1980). Periodic Spontaneous Collapse and Revival in a Simple Quantum Model. *Physical Review Letters*, *44*(20), 1323–1326. https://doi.org/10.1103/physrevlett.44.1323

Karnieli, A., & Fan, S. (2023). Jaynes-Cummings interaction between low-energy free electrons and cavity photons. *Science Advances*, *9*(22). https://doi.org/10.1126/sciadv.adh2425

C. Di Fidio, Vogel, W., M. Khanbekyan, & Welsch, D.-G. (2008). Photon emission by an atom in a lossy cavity. *Physical Review. A, Atomic, Molecular, and Optical Physics/Physical Review, A, Atomic, Molecular, and Optical Physics*, *77*(4). https://doi.org/10.1103/physreva.77.043822

Mang, F., XiMing, F., Lei, S., XiWen, Z., & Min, Y. (1999). Approximate Solution to the Driven Jaynes–Cummings Model with Small Detunings. *Communications in Theoretical Physics*, *32*(2), 181–188. https://doi.org/10.1088/0253-6102/32/2/181

Manzano, D. (2020). A short introduction to the Lindblad master equation. *AIP Advances*, *10*(2), 025106. https://doi.org/10.1063/1.5115323

Stephen M. Barnett & John Jeffers (2007): The damped Jaynes–Cummings model, Journal of Modern Optics, 54:13-15, 2033-2048. http://dx.doi.org/10.1080/09500340701352813

Lodahl, P., Floris van Driel, A., Nikolaev, I. S., Irman, A., Overgaag, K., Vanmaekelbergh, D., & Vos, W. L. (2004). Controlling the dynamics of spontaneous emission from quantum dots by photonic crystals. *Nature*, *430*(7000), 654–657. https://doi.org/10.1038/nature02772

Viola, L., & Lloyd, S. (1998). Dynamical suppression of decoherence in two-state quantum systems. *Physical Review A*, *58*(4), 2733–2744. https://doi.org/10.1103/physreva.58.2733

Raimond, J.-M. (2014). Exploring the quantum world with photons trapped in cavities and Rydberg atoms. *Oxford University Press EBooks*, 395–426. https://doi.org/10.1093/acprof:oso/9780199681181.003.0011

APPENDIX

Appendix A : Program Code.

A1 : All Python/Matlab Programs used.

Click here

Appendix B : Tedious derivations

B1 : Derivation of Lindblad master equation can be obtained from a paper by (C. A. Brasil, 2012) here

B2 : Master equation for cavity decay (Was done in scientific workplace)

For cavity decay (photon loss), we need to find the matrix element one-by-one :

$$\dot{\rho}(t) = -\frac{i}{\hbar}[\hat{H},\rho] + \kappa\left(a\rho a^{+} - \frac{1}{2}\{a^{+}a,\rho\}\right)$$

where k is cavity decay. The first term :

$$< g,m|[\hat{H},\rho]|g,n >= (m-n)\hbar\omega_L < g,m|\rho|g,n > +\hbar\lambda(< e,m-1|\rho|g,n > - < g,m|\rho|e,n-1 >)$$

$$< e,m|[\hat{H},\rho]|e,n >= (m-n)\hbar\omega_L < e,m|\rho|e,n > +\hbar\lambda(< g,m-1|\rho|e,n > - < e,m|\rho|g,n-1 >)$$

$$< e,m|[\hat{H},\rho]|g,n >= \hbar(2\omega_0 + (m-n)\omega_L) < e,m|\rho|g,n > +\hbar\lambda(< g,m+1|\rho|g,n > - < e,m|\rho|e,n-1 >)$$

$$< g,m|[\hat{H},\rho]|e,n >= \hbar(-2\omega_0 + (m-n)\omega_L) < g,m|\rho|e,n > +\hbar\lambda(< e,m-1|\rho|e,n > - < g,m|\rho|g,n+1 >)$$

The second term :

For gg :

$$< g,m|a\rho a^{+}|g,n >= \sqrt{(m+1)(n+1)} < g,m+1|\rho|g,n+1 >$$

$$< g,m|a^{+}a\rho|g,n >= m < g,m|\rho|g,n >$$

$$< g,m|\rho a^{+}a|g,n >= n < g,m|\rho|g,n >$$

For ee :

$$< e,m|a\rho a^{+}|e,n >= \sqrt{(m+1)(n+1)} < e,m+1|\rho|e,n+1 >$$

$$< e,m|a^{+}a\rho|e,n >= m < e,m|\rho|e,n >$$

$$< e,m|\rho a^{+}a|e,n >= n < e,m|\rho|e,n >$$

For ge :

$$< g,m|a\rho a^{+}|e,n >= \sqrt{(m+1)(n+1)} < g,m+1|\rho|e,n+1 >$$

$$< g,m|a^{+}a\rho|e,n >= m < g,m|\rho|e,n >$$

$$< g,m|\rho a^{+}a|e,n >= n < g,m|\rho|e,n >$$

For eg :

$$< e,m|a\rho a^{+}|g,n >= \sqrt{(m+1)(n+1)} < e,m+1|\rho|g,n+1 >$$

$$< e,m|a^{+}a\rho|g,n >= m < e,m|\rho|g,n >$$

$$< e,m|\rho a^{+}a|g,n >= n < e,m|\rho|g,n >$$

Summing them up (1st and 2nd terms) :

$$\dot{\rho}_{gg}^{mn} = \ < g,m|\dot{\rho}|g,n > = -i[(m-n)\omega_L < g,m|\rho|g,n > + \lambda(\sqrt{m} \ < e,m-1|\rho|g,n > - \sqrt{n} \ < g,m|\rho|e,n-1 >)] + \kappa[\sqrt{(m+1)(n+1)} \ < g,m+1|\rho|g,n+1 > - \frac{(m+n)}{2} < g,m|\rho|g,n >]$$

$$\dot{\rho}_{ge}^{mn} = \ < g,m|\dot{\rho}|e,n > = -i[(-\omega_0 + (m-n)\omega_L) < g,m|\rho|e,n > + \lambda\left(\sqrt{m} \ < e,m-1|\rho|e,n > - \sqrt{n+1} \ < g,m|\rho|g,n+1 >\right)] + \kappa[\sqrt{(m+1)(n+1)} < g,m+1|\rho|e,n+1 > - \frac{(m+n)}{2} < g,m|\rho|e,n >]$$

$$\dot{\rho}_{eg}^{mn} = \ < e,m|\dot{\rho}|g,n > = -i[(\omega_0 + (m-n)\omega_L) < e,m|\rho|g,n > + \lambda\left(\sqrt{m+1} \ < g,m+1|\rho|g,n > - \sqrt{n} \ < e,m|\rho|e,n-1 >\right)] + \kappa[\sqrt{(m+1)(n+1)} < e,m+1|\rho|g,n+1 > - \frac{(m+n)}{2} < e,m|\rho|g,n >]$$

$$\dot{\rho}_{ee}^{mn} = \ < e,m|\dot{\rho}|e,n > = -i[(m-n)\omega_L < e,m|\rho|e,n > + \lambda\left(\sqrt{m+1} \ < g,m+1|\rho|e,n > - \sqrt{n+1} \ < e,m|\rho|g,n-1 >\right)] + \kappa[\sqrt{(m+1)(n+1)} < e,m+1|\rho|e,n+1 > - \frac{(m+n)}{2} < e,m|\rho|e,n >]$$